

RegLIC: Supplementary Methods and Controlled Experiments

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On this fixed cohort the router-minus-dither MAC margin is 2.10, 2.40, 1.28, 0.32, 0, and 0 percentage points as the nominal target moves from 0.05 to 0.50 dB (Fig. S2). Its paired BD-proxy differences are -0.050 , -0.085 , -0.036 , -0.015 , 0, and 0 points; the 0.05 dB 95% interval includes zero. At the two loose targets all four policies reach the same archived maps on these 25 sequences. Using every five-QP-complete sequence separately at each target instead yields 25/51/53/53/53/53 sequences and router-minus-dither MAC margins of 2.10/2.57/2.02/1.31/0.38/0.02 points. The fixed-cohort panel prevents changes in sequence membership from masquerading as a budget effect; the per-budget cohort retains the broader 0.10 dB result. The separate depth-bank Kodak BD-rate estimates use independent networks and estimated entropy rates and must not be pooled with this shared-latent BD proxy. The exact convolution recount lowers the complete-cohort 0.10 dB router and dither savings from the older normalized model’s 28.22/25.63% to 27.54/24.97%, while preserving a 2.57-point margin. Earlier MAC figures in the manuscript retain that normalized model; this audit uses padded-frame convolution counts and is the more precise arithmetic comparison, still not measured runtime.

S1. Scope and Reproduction

This supplement records CPU analyses of the frozen evaluation archive, actual-byte evaluation of frozen interim depth models, and the protocol for the ongoing full-schedule depth study. The main manuscript uses the recorded mixed reconstructions of the shared-latent e15 codec. Control-path enumeration and the analytic permutation diagnostic reuse stored source-error tables without executing the codec. The fixed-control, released-reference and component-intervention sections instead report new CPU reconstructions. The separate GPU-kernel section measures one routed synthesis case, not complete-codec latency. Independent D2/D4/D6 training has completed; D8/D10/scratch D12 remain in progress. Interim measurements are labelled and do not stand in for completed full-schedule results.

The stand-alone paper repository contains the plotted source tables, source hash manifests, ed-

itable figure scripts and inspection results under `data/refresh20260927`. Figures can be regenerated on CPU using `scripts/build_figures.py` and `scripts/build_extended_figures.py`. The parent research repository also retains the scripts that derive these summaries from the raw records and checkpoints. Hashes identify the files inspected during this revision; they do not retrospectively establish the exact executable used by a historical run whose code hash was not recorded.

S2. Tracing the Reconstruction Reference

Two measurement paths need to be distinguished. The source-table producer loads the e15 network and a separate released-weight reference. It computes the candidate tile errors $M_{t,k}$ from e15 and the scalar reference error R from the released synthesis network, both on the padded source. The final-reconstruction evaluator instead loads e15 alone, calls its full-frame synthesis path, crops both full and mixed outputs, and measures their difference in source quality. Its raw field named `psnr_release` is consequently a fine-tuned full-depth reference in the inspected code (Fig. S5).

Metric provenance. The legacy field `db_rgb` is also misnamed. The CTC reader upsamples YCbCr 4:2:0 to 4:4:4 and centres the three planes. Synthesis returns this representation, and the evaluator computes equal-channel tensor MSE directly, before RGB conversion or clipping. Both the padded selection tables and the cropped MSE-ratio loss therefore use YCbCr 4:4:4. We label this quantity Δ_{444} or “444-MSE loss” throughout. The separately recorded 6:1:1 metric is a weighted average of plane PSNRs after chroma downsampling and clipping, not the same MSE ratio.

Two fixed-map CPU replays with strict checkpoint loading reproduce the archived `db_rgb` values within 1.1×10^{-6} dB (Fig. S3). Converting to RGB changes the result. These checks establish the interpretation of the inspected path; they are not a new 265-pair RGB evaluation. Legacy JSON keys and historical annotations are preserved for traceability; `METRICS.md` records their corrected meaning and the replay hashes. The separate independent-depth validation explicitly converts reconstructions to RGB and must not

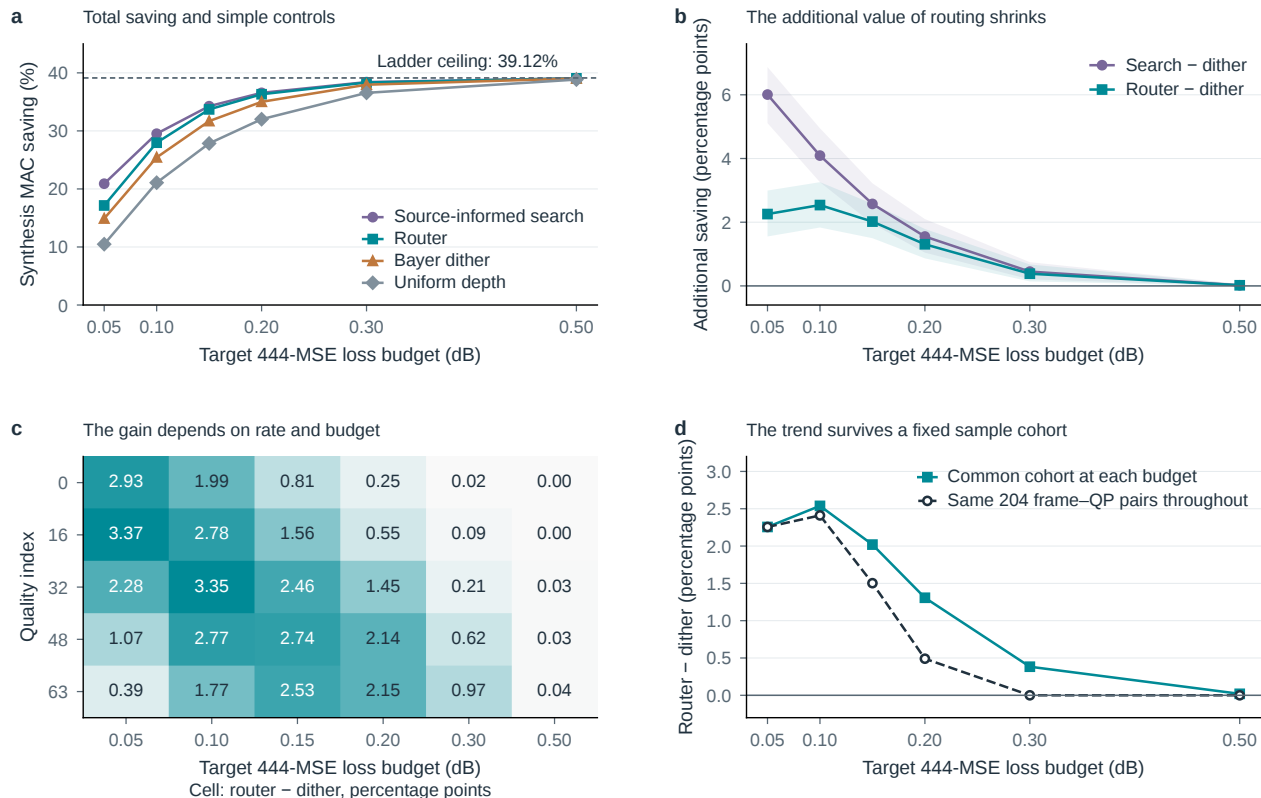


Figure S1. **Budget sensitivity beyond the compact main comparison.** **a,b**, Total saving and paired increments over dithering. **c**, The router increment stratified by QP. **d**, Repeating the comparison on the same 204 frame-QP pairs preserves the contracting margin. Intervals resample sequences and keep QPs together. All policies use per-source calibration; nominal budgets are table targets, not final-image guarantees.

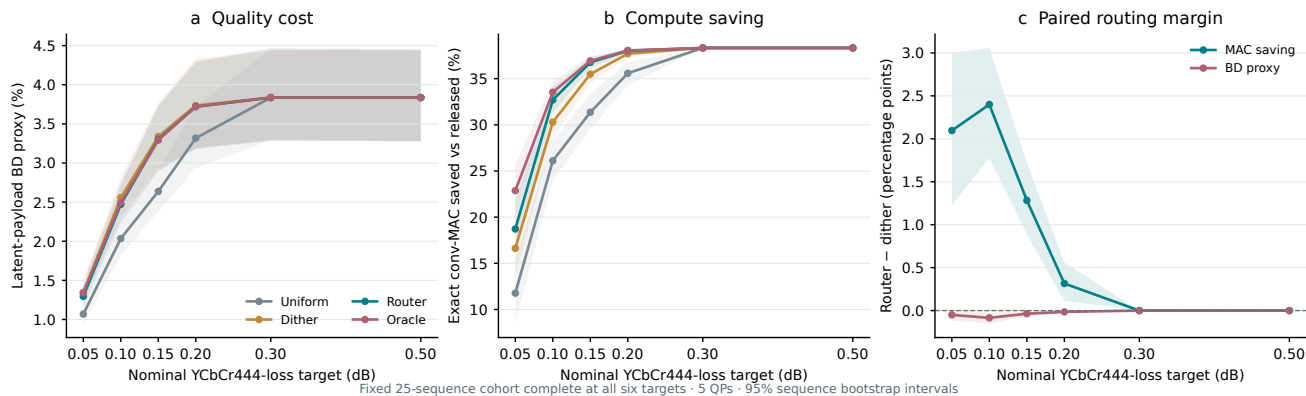


Figure S2. **Rate-equivalent cost and compute saving across six nominal early-exit budgets.** The same 25 CTC first-frame sequences have all five QPs and all four policies at every target. **a**, Sequence-mean YUV 6:1:1 BD-rate *proxy* against dense e15, using actual round-tripped latent payload bytes but excluding exit-map and container bits. **b**, Exact convolution MAC saving against released D12 full synthesis on the padded frame, including adapters and seam repair but excluding control and execution overhead. **c**, Paired router-minus-dither margins with 95% sequence-cluster bootstrap bands. Policies use source-informed calibration; neither the budget nor the BD proxy establishes deployable full-codec performance.

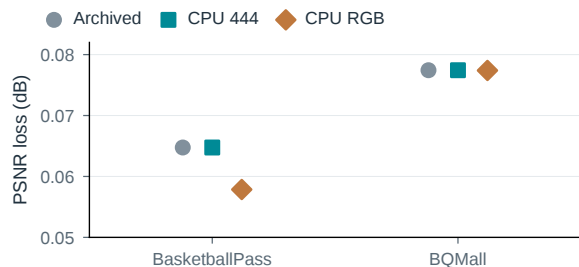


Figure S3. **The archived field measures 444 loss, not RGB loss.** Two QP32 router maps at the nominal 0.1 dB target. The archived value agrees with CPU unclipped YCbCr 4:4:4 replay; explicit clipping and RGB conversion change the result. Losses are relative to e15 full-frame. These are two metric checks.

be pooled with this legacy metric.

A CPU comparison of the two state dictionaries finds 255 shared encoder/entropy tensors with no changes. All 143 inherited synthesis tensors that can be compared have changed, with a maximum absolute parameter difference of 0.2481. Thus the reporting path cannot be identified with released synthesis merely because it executes all twelve trunk blocks. After exact key remapping, all 398 inherited warm-start tensors also match the archived official D12 release. Extra adapter/router tensors are outside this comparison. This second check ties the inspected warm start to the released parameters without assuming that its filename proves provenance.

This distinction limits the interpretation of threshold crossings. The recorded 7/263 YCbCr 4:4:4 and 49/263 YUV losses above a nominal 0.1 dB target combine differences in reference, image support, metric and the influence of the shared repair/head. They do not identify the contribution of any single factor. The paired MAC accounting remains attached to the same recorded maps, but the exact quality gap to the cropped released decoder cannot be recovered from the exported JSON alone.

The completed QP32 replay in Sec. S16.1 retains both full-depth anchors and explicitly computes valid-pixel RGB errors on all 53 frames. It closes this reference gap at that QP without relabelling the five-QP archive. Selection and final verification must still declare the same reference and metric; any deployment fallback must include its bytes and runtime.

S3. What Additional Synthesis Depth Buys

Each column of the archived source-error table was produced by decoding a uniform-depth map and measuring per-tile source MSE. Its equal-area tile average is therefore the

padded YCbCr 4:4:4 MSE of that actual uniform reconstruction. Comparing this mean to the separately evaluated released reference gives mean losses of 0.2045, 0.0968, 0.0605 and 0.0379 dB for 6/8/10/12 blocks over all 265 pairs. The corresponding modelled MAC savings are 39.12%, 24.22%, 13.03% and -0.95% . The negative all-deep saving reflects shared repair overhead relative to the full-frame reference. These values use padded support and cannot substitute for cropped released-reference replay.

For adjacent depths, the local quality improvement is $g_t = 10 \log_{10}(M_{t,k}/M_{t,k+1})$. The frame-median improvement, averaged across pairs, is 0.0993 dB for 6-to-8 blocks, 0.0320 dB for 8-to-10, and 0.0213 dB for 10-to-12. The respective mean within-frame interquartile ranges are 0.0748, 0.0340 and 0.0238 dB. This quantifies both diminishing returns and variation within the same image. The average fractions of tiles whose error increases are 4.37%, 11.55% and 11.41%; deeper synthesis is not uniformly better at every location.

Figure S6 uses equal frame-QP weighting, rather than letting larger images dominate through their tile counts. Sequence-cluster intervals retain all five QPs together. Each tile comparison still changes the depth of its neighbours because the underlying maps are uniform. It is a conditional reconstruction diagnostic, not a causal one-tile intervention or a completed independent-codec depth ablation.

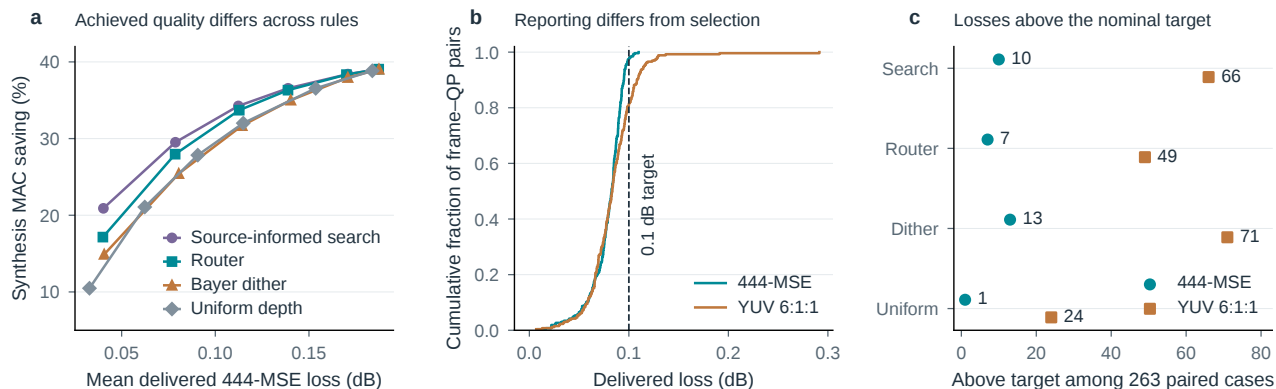
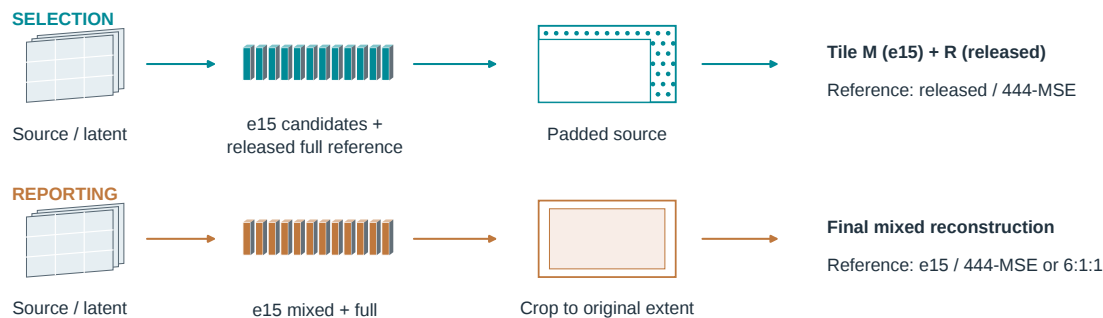


Figure S4. **Nominal targets and final reporting answer different questions.** **a**, Recorded average cropped YCbCr 4:4:4 loss against MAC saving. **b**, Router YCbCr 4:4:4 and YUV 6:1:1 loss distributions at a nominal 0.1 dB target. **c**, Counts above that numerical target, among the 263 common cases and with 10^{-4} dB tolerance. Selection uses a released reference on padded YCbCr 4:4:4; reporting uses cropped e15 full-frame synthesis. The counts therefore do not isolate a common-reference guarantee failure.

a Selection and reporting use different reconstruction anchors and image support



CPU audit: 255 shared encoder/entropy tensors unchanged; 143/143 inherited synthesis tensors changed.

Required comparison: reconstruct both anchors, use the same valid pixels and metric, then verify the selected output.

Figure S5. **Reference and support are part of the measurement.** Candidate tile errors come from e15, whereas the table denominator comes from a separate released-weight reference. Final reporting uses the cropped full-depth e15 reconstruction. The CPU tensor comparison establishes a weight difference in the inspected checkpoints. The figure traces code paths and does not depict newly decoded images.

Quality-index interaction. The value of depth also changes across the five encoded quality indices (Fig. S7). The mean frame-median gain from blocks 6 to 8 rises from 0.0665 dB at QP 0 to 0.1315 dB at QP 63. Its paired QP63-minus-QP0 change is 0.0650 dB [0.0467, 0.0844] over 53 sequences; for blocks 8 to 10 it is 0.0329 dB [0.0247, 0.0416]. The 10-to-12 change is 0.0065 dB [−0.0009, 0.0145], unresolved at this sample size. This is evidence for conditioning a depth decision on QP, but these uniform-map comparisons

cannot quantify the gain of a learned spatial selector or the latency of conditional execution.

Bitstream-level routed decisions and control transfer. Figure S8 follows each entropy-decoded latent through decoder-side routing, assembled reconstruction and seam repair. Equal weighting across the 24 images and five QPs gives 17.16% mean analytical synthesis MAC saving for the CTC-calibrated QP-specific control, with a 24-

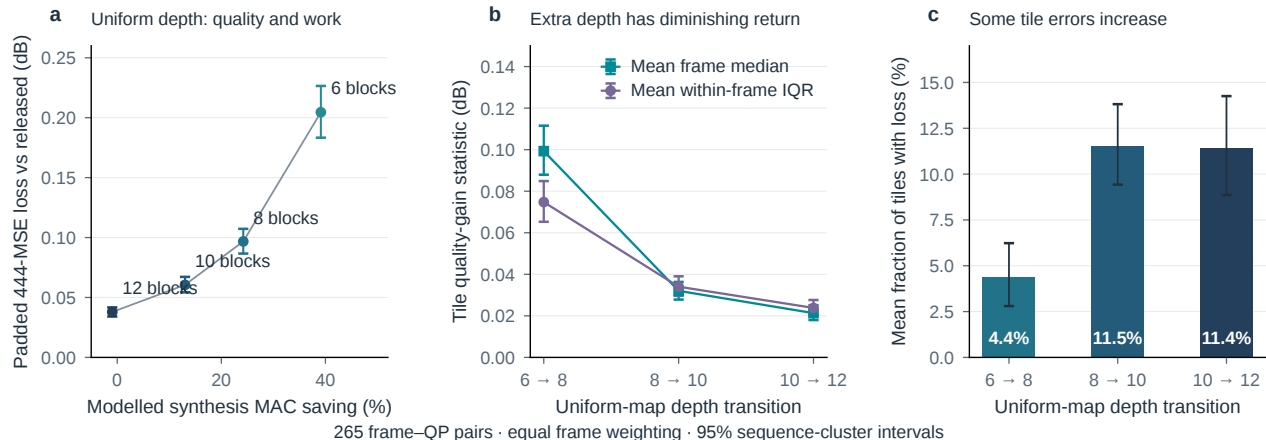


Figure S6. **The value of additional depth varies across regions.** **a**, Uniform-depth padded YCbCr 4:4:4 quality against the released anchor and the recorded MAC model. **b**, Mean frame-median tile improvement and mean within-frame IQR for adjacent depths. **c**, Mean fraction of tiles whose error increases. Error bars are 95% sequence-cluster intervals. These are actual archived uniform reconstructions; neighbouring depths change together, and independent shallow codecs are not represented.

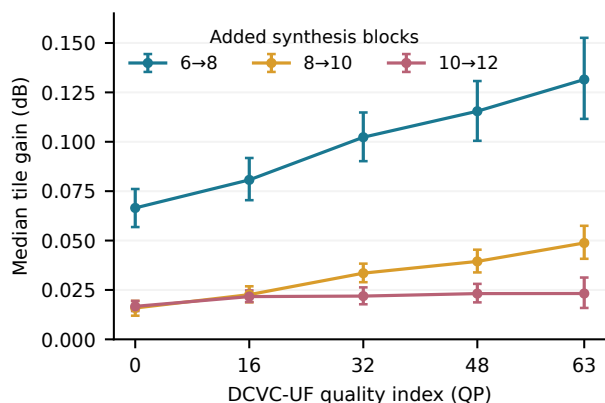


Figure S7. **Marginal synthesis depth depends on quality index.** Each point averages the within-frame median tile gain across 53 sequences for two adjacent uniform-depth outputs on padded YCbCr 4:4:4. Bars are paired 95% sequence-cluster intervals from 5,000 draws. Neighbouring tiles change depth together; these are diagnostic comparisons, not isolated tile interventions or independent codec-bank results.

image cluster-bootstrap interval of [14.81,19.58]%. By QP, saving is 31.88/24.81/11.56/9.29/8.28%; the deepest exit takes 0/0/47.9/59.7/65.3% of tiles. These figures are lower at high QP than the 53-CTC-frame calibration means of 35.27/31.85/25.44/23.25/20.38%.

The rate trend is partly a property of the control: its price β falls from 30.68 at QP0 to -23.52 at QP63. Recomputing decisions from the *same* router scores with β fixed at the QP16 value raises the five-QP mean MAC saving to 26.82%, but actual reconstruction raises mean Δ_{444} from

0.0565 to 0.1223 dB. At QP48/63, the fixed control gives 0.147/0.269 dB instead of 0.047/0.063 dB; 55 of 120 fixed-control cases exceed 0.1 dB, versus 14 with QP-specific control. Thus fixing β does not recover free compute. Holding β constant does not remove QP from the router head or latent statistics, so these measurements do not identify an intrinsic QP effect on capacity demand. Kodak is an external control-transfer check, not a set on which we retune and re-evaluate β . Separate released-D12-relative YUV 6:1:1 PSNR records are archived and must not be read as this Δ_{444} target.

S4. Independent-Image Calibration of the Decoder-Side Control

We next fit the scalar price without using Kodak records in the optimisation. The 100 original DIV2K validation PNGs are ordered by the SHA256 of their filenames. The first 24 supply calibration; the next, disjoint 24 supply an internal transfer check. Every image contributes a central 768×512 RGB crop, encoded into actual FUFREF2 streams at QP 0/16/32/48/63. The released analysis and entropy tensors are shared with the e15 model. The same decoder-side stem-QP router, six-tile assembly, seam repair, full-frame e15 reconstruction and cropped YCbCr 4:4:4 metric are used at every candidate price. This is an untimed, single-core CPU replay.

The candidate grid, fixed before evaluation, contains prices from -50 to 60 in increments of five, plus the exact earlier CTC price at each QP and the exact QP16 price. For each QP separately, the primary rule maximises mean analytical synthesis MAC saving among candidates whose mean Δ_{444} on the 24 calibration images

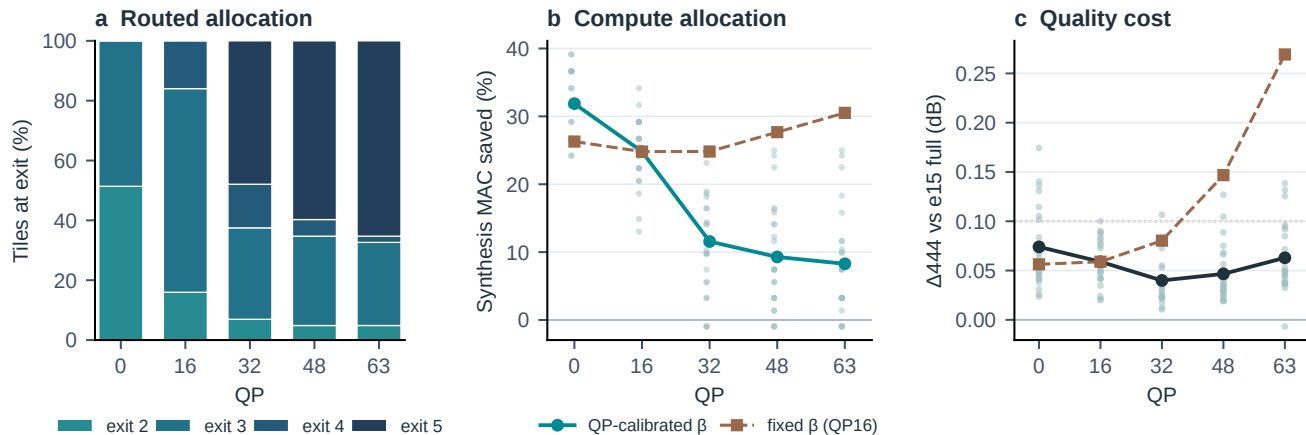


Figure S8. **QP-dependent routing on the complete Kodak set.** All 24 images at QP 0/16/32/48/63 are encoded into 120 FUFREF2 research bitstreams and independently decoded before the same stem-QP router selects six 256×256 tiles per image. **a**, pooled calibrated exit shares; **b**, analytical synthesis MAC saving relative to released D12 for the QP-calibrated control and the counterfactual that holds β at its QP16 value; **c**, actual assembled-image Δ_{444} against e15 full-frame output for both controls. Points show images and lines show 24-image means. The dotted 0.1 dB line is the nominal target, not a per-image guarantee. MAC accounting includes adapters and seam repair but excludes router work; it is not measured latency. FUFREF2 is not the native CUDA wire format.

is at most 0.1 dB. It selects $\beta = (40, 15, 15, 10, -10)$ at increasing QP. Mean saving/loss is 20.73%/0.0851 dB on calibration, 20.05%/0.0889 dB on held-out DIV2K, and 25.88%/0.0976 dB on Kodak24 $\times$ QP5. On Kodak, the 24-image cluster-bootstrap 95% intervals are [23.29, 28.21]% MAC saving and [0.0832, 0.1128] dB mean Δ_{444} . The earlier CTC and fixed-QP16 controls, evaluated on the same Kodak streams, yield 17.16%/0.0565 dB and 26.82%/0.1223 dB, respectively.

The pooled target is not a hard guarantee. Under the primary policy, 47/120 Kodak cases exceed 0.1 dB; QP48/63 means are 0.1275/0.1261 dB, with 17/24 and 15/24 cases above the threshold. The largest Kodak loss is 0.2989 dB at QP63. The worst DIV2K calibration loss, 0.7918 dB at QP0, was recomputed with the direct, uncached decoder and matched the cached candidate exactly. These tails motivate a risk-aware control rather than an unqualified quality-preservation claim.

After inspecting the calibration aggregate, we also explored a less conservative rule: maximise five-QP mean saving subject only to five-QP mean calibration loss ≤ 0.1 dB. Its price vector (50, 15, 15, 14.0229, 5) reaches 23.09%/0.09994 dB on calibration, but 23.07%/0.1131 dB on held-out DIV2K and 28.64%/0.1228 dB on Kodak. Thus the apparent target hit does not transfer. The rule was selected using calibration alone and committed before Kodak replay, but it was introduced after inspecting that cohort and is therefore labelled exploratory. Kodak had been examined in prior diagnostics; it is independent of these specific price

fits, not globally untouched. A fresh cohort is needed for a final Q90 or maximum-loss-constrained policy comparison.

The router is not the only source of tail loss. On the 120 DIV2K calibration cases, forcing every tile to the deepest exit still exceeds 0.1 dB in five cases: every QP of image 0828. Its all-deep tiled losses are 0.115/0.164/0.152/0.144/0.145 dB at increasing QP. No scalar price on the frozen candidate grid can meet the threshold for those cases. This is a tiled-versus-full-frame floor, not proof that every possible spatial map or decoder architecture has the same floor. Direct uncached replay exactly reproduces the QP0 primary-map loss of 0.7918 dB and all-deep floor of 0.1148 dB. A 16-pixel band at the internal tile boundaries occupies 13.64% of pixels but contains 72.43% of the all-deep/full-frame squared difference energy; its mean difference is 16.6 times the interior. The structural floor and the router-dependent excess therefore require separate controls (Fig. S10).

An outcome-informed oracle restricted to the same finite scalar-price grid reduces the calibration threshold failures from 34 to those five cases, while saving 21.60% mean synthesis MAC versus the fixed primary policy’s 20.73%. It consults source reconstruction quality and is not deployable. Of the primary policy’s 34 calibration violations, 29 are below threshold under tiled all-deep decoding; an ideal source-informed binary fallback would remove those 29 but lower mean saving to 13.48%. The candidate-grid oracle retains 21.60% because it chooses cheaper feasible maps

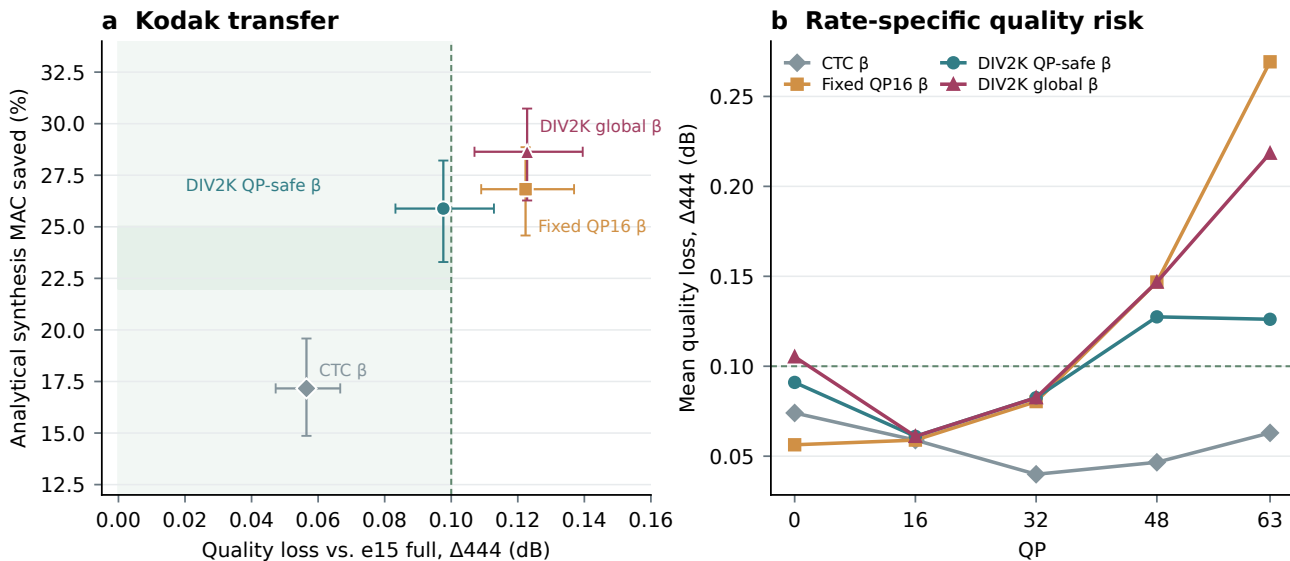


Figure S9. **A fixed price transfers in the mean, but leaves a quality tail.** **a**, Equal-image-and-QP mean Kodak operating points from actual assembled reconstructions on the same 120 bitstreams; bars are 95% bootstrap intervals over 24 images. The green region marks mean $\Delta_{444} \leq 0.1$ dB and 22–25% analytical synthesis MAC saving. **b**, Rate-specific mean loss exposes the QP48/63 failures hidden by the pooled mean. The global-QP price rule is exploratory and was frozen before Kodak transfer. MAC accounting excludes router/entropy work and is not measured latency.

rather than always sending a risky frame to all-deep. This gap is a motivation for a decoder-available risk signal, not a measured router result. As a first negative control, we fitted a fixed L2 logistic risk score from decoder-visible QP and six-tile exit-map summaries on the 24 calibration images, then evaluated once on the disjoint 24-image DIV2K set. Its violation AUC drops from 0.785 to 0.646, slightly below the 0.651 AUC of mean exit depth alone. A calibration-locked threshold flags 74/120 validation cases yet misses 7/35 violations. Map geometry alone is not a reliable quality guard in this pilot; additional decoder-available evidence and a fresh cohort are required before proposing a fallback policy. The unchanged score ranks Kodak24 $\times$ QP5 cases better (AUC 0.779), but the same threshold flags 97/120 cases to catch 44/47 violations. Kodak was not used in this fit, although it had been inspected in earlier project diagnostics; this is a transfer check, not a pristine external holdout. Conversely, a fixed per-QP Q90 calibration rule reduces mean saving to 11.20% on these calibration cases. These are explanatory calibration ablations, not external transfer results. They show why simply tightening the scalar price may sacrifice much of the compute gain without repairing the tiled reconstruction floor.

A context counterfactual separates the two failure mechanisms. For image 0828 we replayed the same five research bitstreams with the fixed router maps and two *diagnostic* suf-

fix implementations: clipped feature windows wide enough to supply every depthwise layer, and a canvas that shares only one-pixel context at each depthwise layer. Both disable the originally trained seam-repair module. With all tiles at the deepest exit, each produces the full-frame e15 output exactly at every QP (maximum absolute difference zero). Thus the tiled all-deep floor is removable by restoring true context; it is not an inherent penalty of the checkpoint. For the frozen mixed map, QP0 still loses 0.720 dB with exact context because its selected exits are mostly shallow. At QP16–63, the same intervention reduces the 0.153–0.172 dB deployed losses to 0.011–0.016 dB, isolating the tile boundary as the dominant cause (Fig. S11).

Exact windows compute the entire costly suffix over expanded areas. Their analytical synthesis cost is 1.315 times full-frame e15 for the QP16–63 mixed maps, compared with 0.968 for the deployed zero-halo path; neither number measures latency. Cropping one invalid context ring after each suffix block is output-identical to the full-window method on 0828/QP63, and lowers its mixed-map synthesis cost to 1.151 times full e15; this still exceeds the full decode. An optimistic union-of-required-cells estimate, which shares all overlapping tile context at zero scheduling cost, is 0.965 times full e15 on that same map: exact quality leaves only 3.5% ideal synthesis saving there. This is an area calculation, not a kernel or latency result. Depthwise-only coupling gives 0.056–0.099 dB on these four QPs without haloing pointwise layers, but the mixed-depth boundary still differs from a true

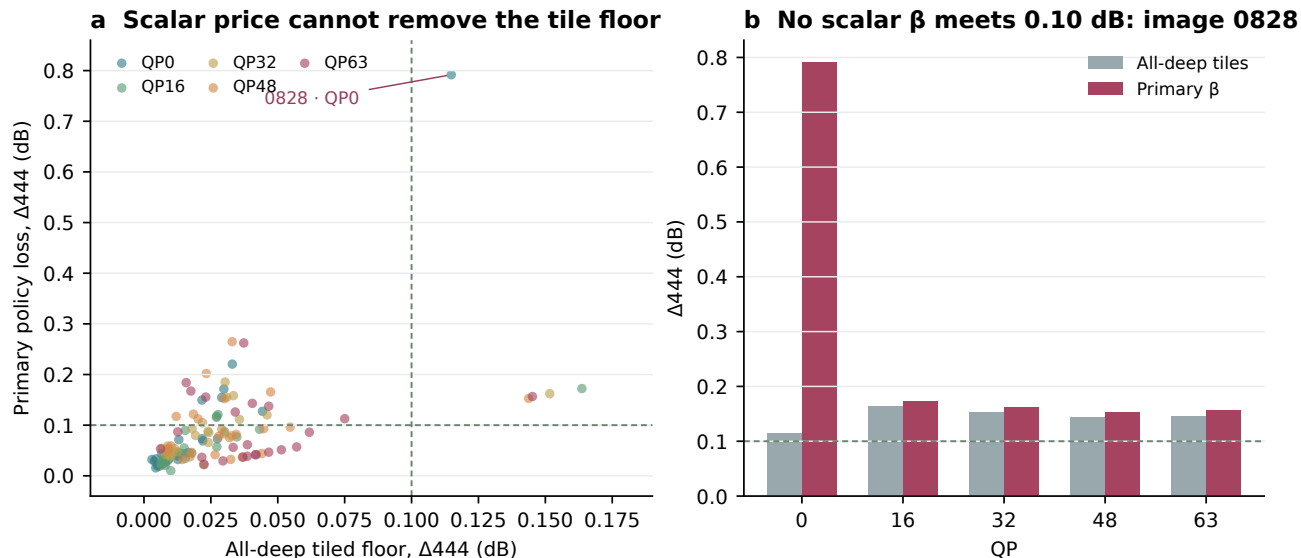


Figure S10. **Separate the routing tail from the tiled reference floor.** **a**, Primary-policy loss against all-deep tiled loss on the same 24 DIV2K calibration images at five QPs; dashed lines mark 0.1 dB. **b**, Image 0828 is above the threshold even when every tile takes the deepest exit. The large QP0 gap between bars is additionally caused by the chosen spatial exit map. Both bars compare with full-frame e15 output; no latency or new trained model is inferred.

full-depth neighbour. Its PyTorch implementation also assembles and gathers halo buffers, so low arithmetic overhead is not evidence of low wall-clock overhead. Retaining the old seam-repair weights can worsen the coupled result (QP63: 0.149 versus 0.099 dB); a coupled decoder is a new variant requiring broader quality and matched latency tests. Indeed, two post-hoc validation stress cases give a counterexample to deploying coupling as a global switch. On 0898/QP48, four of six tiles are deepest, yet coupling without repair raises loss from 0.165 to 0.365 dB against an otherwise matched zero-padding/no-repair control. On the shallow 0844/QP48 map it barely changes loss (0.573 to 0.572 dB). These cases were selected after inspecting validation failures; they establish a failure mode, not its frequency. The mixed-depth neighbour state needs a new boundary rule or separately validated fallback.

The single outlier is not the cohort's compute profile.

Applying the same optimistic union-of-required-cells calculation to all 240 frozen maps in the 24-image calibration and 24-image disjoint DIV2K cohorts gives 21.30% and 20.55% mean *ideal* synthesis MAC saving. On validation the QP0/QP63 means are 33.45%/9.00%, respectively (Fig. S12). The 85 maps already within 0.1 dB under the deployed zero-halo decoder retain 18.61% mean ideal saving; the 35 violating maps retain 25.28%. This grouping uses *deployed* quality; the mask geometry alone does not establish quality or latency.

Exact context recovers quality, but not every shallow

exit. We next reconstructed all 120 real bitstreams in the disjoint 24-image validation cohort under the same locked route maps, replacing the deployed zero-halo-plus-repair suffix with clipped sufficient-context windows and no repair. Mean loss relative to full-frame e15 falls from 0.0889 to 0.0681 dB; the paired improvement is 0.0208 dB (24-image cluster-bootstrap 95% interval 0.0155–0.0266 dB). The count above 0.1 dB falls from 35 to 24: 11 cases are rescued and no new threshold violation appears. Two QP63 cases worsen slightly (at most 0.0035 dB), consistent with the old learned repair occasionally helping. The 24 persistent violations are largely a depth-selection problem; e.g., image 0844/QP48 remains 0.465 dB behind full-frame e15 with route [2, 2, 2, 2, 3, 3]. The separate beta-fitting 24-image calibration partition shows a similar 0.0210 dB mean recovery and 34-to-24 drop in violations; it is a mechanism check, not an independent evaluation set.

For these maps the ideal union-of-required-cells model gives 20.55% mean synthesis MAC saving; 67/120 cases satisfy both exact-context loss ≤ 0.1 dB and ideal saving $\geq 10\%$. These are *potential* operating points, not a latency result: the direct per-tile exact-window code recomputes overlaps, while an efficient shared sparse implementation has not been built. All comparisons here use e15 full-frame centred YCbCr444, not released-decoder YUV611 or full-codec BD-rate.

A source-informed diagnostic clarifies the depth headroom in the 24 persistent violations: uniformly raising all

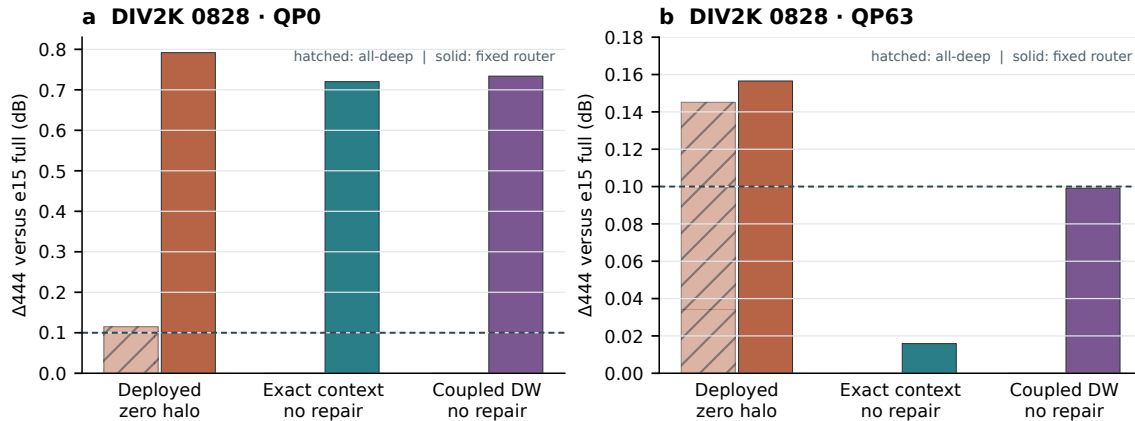


Figure S11. **A targeted context ablation on the DIV2K 0828 outlier.** The hatched bars force all six tiles to the deepest exit; solid bars retain the frozen primary router map. **a**, At QP0, supplying exact context removes the all-deep seam floor but leaves a large mixed-map loss from early exit. **b**, At QP63, the fixed map is mostly deepest and tile context is the main loss. The same five-QP numerical audit is archived; this single-image counterfactual does not establish cohort-level performance or decoder latency. All losses use e15 full-frame YCbCr444 as reference.

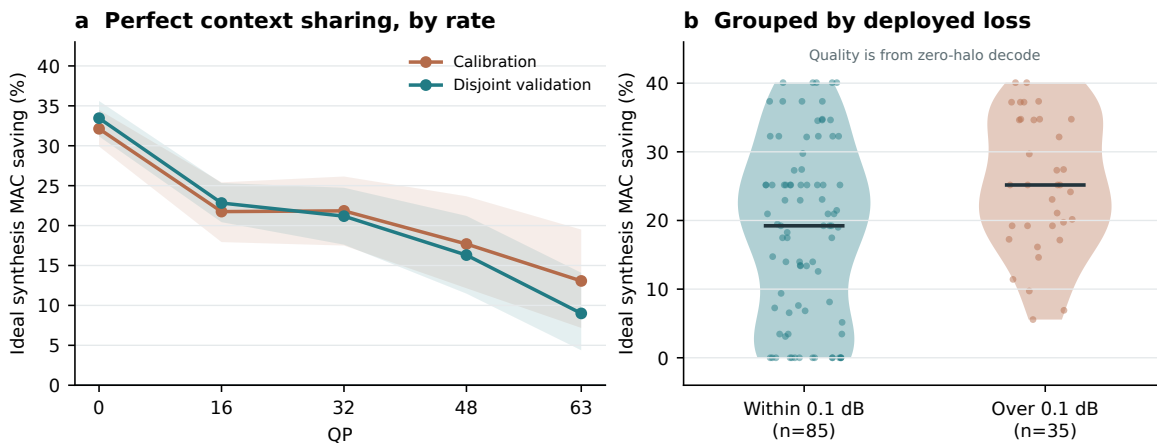


Figure S12. **Optimistic exact-context arithmetic across fixed maps.** **a**, Union-of-required-feature-cell MAC saving by QP, with 24-image cluster-bootstrap 95% intervals for the disjoint calibration/validation sets. **b**, Validation maps grouped by *deployed zero-halo* loss against e15 full-frame. These are geometry and cost calculations; the quality grouping is from the deployed path; the exact-context quality measurement on the disjoint validation cohort follows separately. No sparse-kernel latency was measured.

364 six tile exits by one group recovers 22 cases below 0.1 dB, and two more require a second group. If one could identify
 365 those cases from decoder-available evidence, this fallback
 366 would leave 17.89% mean *ideal* cohort MAC saving, versus
 367 14.96% from sending every violating case to all-deep. These numbers use ground-truth image error to choose when
 368 to stop, so they are *oracle bounds*, not a trained router, a prospective held-out result, or measured runtime. They motivate
 369 a calibrated risk signal and targeted capacity increase rather than a blanket all-deep fallback.

370 A finer source-informed counterfactual advances only
 371 selected tiles by one exit group. Eight of the 24 failures
 372
 373

376 cross the cap after one tile, and a further 14 after two to
 377 six tiles; 0844/QP32 and 0844/QP48 require the uniform
 378 two-group increment. Choosing the fewest upgraded tiles
 379 and then the highest ideal saving among ties yields 19.05%
 380 mean ideal saving across all 120 cases, versus 17.89% for
 381 the uniform fallback. This 1.17-point gap measures potential
 382 spatial targeting headroom under source feedback; it does
 383 not establish a decoder-visible tile selector or sparse-kernel
 384 latency. It also does not optimize arbitrary multi-group tile
 385 assignments.

386 We tested whether inexpensive latent statistics provide
 387 that signal for the *exact-context* failure label. A fixed L2

logistic model fitted on calibration QP and map geometry reaches 0.755 AUC on the disjoint validation images; adding stream length, latent moments, spatial variation, and tile-aligned energy gives 0.760 AUC (paired image-bootstrap difference 95% interval $[-0.088, 0.080]$). At calibration-locked thresholds the expanded model misses 11/24 validation violations while flagging 37/120 cases; the geometry-only model misses 9/24 while flagging 43/120. The richer descriptor improves calibration fit but does not establish a transferable guard. This validation set had already informed the mechanism analysis, so the comparison is exploratory; a new prospective cohort is needed before any risk-gated deployment claim.

The context effect transfers to Kodak, but most released gap remains. We replayed all 24 Kodak images at five QPs from their archived real FUFREF2 streams, with the same DIV2K-locked beta and unchanged route maps. The deployed and diagnostic exact-context YUV611 PSNRs average 34.8879 and 34.9053 dB, respectively, against 35.0261 dB from released D12 on the same streams. Exact context recovers 0.0174 dB (24-image paired bootstrap 95% interval 0.0140–0.0209 dB) and lowers the mean gap to released from 0.1382 to 0.1208 dB. Relative to full-frame e15, the count exceeding 0.1 dB Δ_{444} falls from 47/120 to 34/120. An independently replayed full-frame e15 output decomposes the original 0.1382 dB mean released-to-deployed gap exactly into 0.0194 dB from released versus e15 training, 0.1014 dB from the unchanged route under exact context, and 0.0174 dB from the deployed tile-context/repair path. Thus routing accounts for 73.4% of the *mean* gap, although the signed shares vary by image and QP. The checkpoint component rises from nearly zero at QP0 to 0.0485 dB at QP63; the route term remains the largest mean component at every QP. Two image/QP pairs lose YUV611 quality, so this is an average effect rather than a per-case guarantee. The ideal shared-context synthesis arithmetic saves 26.30% on these maps, but no sparse runtime has been measured. Kodak had appeared in earlier project diagnostics; this replay is an external-dataset mechanism check, not a pristine prospective test (Fig. S14).

S5. Auditing the Scalar Control Search

For router scores $\ell_{t,k}$ and costs c_k , increasing the price β in $k_t(\beta) = \arg \max_k (\ell_{t,k} - \beta c_k)$ cannot increase selected cost. To see this, write the optimality inequalities at $\beta_1 < \beta_2$ for their respective choices k_1, k_2 and add them. The result is

$$(\beta_2 - \beta_1)(c_{k_1} - c_{k_2}) \geq 0. \quad (\text{S1})$$

No corresponding statement follows for source error $M_{t,k}$, because that quantity is absent from the score being optimised. A feasibility test at the most expensive endpoint can

therefore miss a feasible middle assignment, and distortion-based bisection can miss a later feasible interval.

We enumerate pairwise score-line intersections, representatives of the open intervals between them, and the endpoints of the archived range $[-20000, 20000]$. Archived controls are included explicitly to retain their floating-point tie behaviour. Identical maps are merged. Among feasible maps, the selected candidate maximises modelled MAC saving, with table loss as a tie-breaker. This is a finite-path audit in float64, rather than an optimiser over all possible tile assignments.

The cost is monotone on every inspected router path, while source loss has at least one reversal on 227/265 paths. At 0.05 dB, enumeration improves 3 of 204 previously feasible plans, by 0.1626 percentage points on average, and recovers one additional feasible case. At 0.1 dB it improves 2 of 263 plans, by 0.0277 points on average. No paired saving changes at the four larger targets. The largest missed saving is 30.45 points at 0.05 dB, so a small mean difference should not be read as a uniform per-frame bound.

We strengthen the blind baseline in the same way. For ordered dithering, all changes occur at a finite set of Bayer-rank thresholds and rung boundaries. Enumerating those maps reveals source-loss reversals on 232/265 paths. It improves 3/204 paired plans at 0.05 dB and 1/263 at 0.1 dB, with mean gains of 0.0192 and 0.0028 points, respectively. Six additional cases become feasible at 0.05 dB. The quality of these newly selected maps has not been measured after final mixed reconstruction, so they do not replace the main manuscript’s reconstruction curves.

The audit reproduces every archived feasible control and cost, verifies cost monotonicity, and checks that enumeration never discards a better archived feasible saving. An independent three-exit example verifies the non-monotone failure mechanism against a dense control grid.

S6. Sequence-Disjoint Control Calibration

The source-calibrated sweep answers which allocation a policy can attain when its scalar control is tuned for each image. It does not establish what happens when that control is fixed before seeing a new source image. We therefore add a CPU diagnostic that keeps the router checkpoint and archived tile tables fixed while separating control calibration by sequence. All five QPs of a sequence stay in one fold. Sorting sequence-name hashes and assigning ranks modulo five yields folds of 10/10/11/11/11 sequences. For each QP, controls are fitted on four folds and evaluated on the fifth; each of the 265 frame–QP pairs is evaluated exactly once per setting. The data have previously informed development, so this is a calibration diagnostic, not an untouched external test.

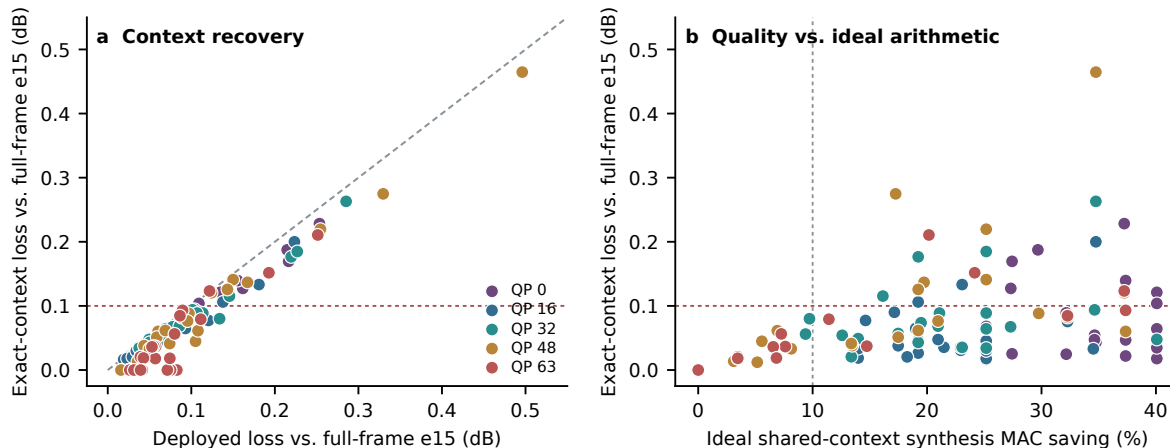


Figure S13. **Exact-context quality on all 120 disjoint DIV2K validation bitstreams.** **a**, Deployed zero-halo-plus-repair versus diagnostic exact-context-without-repair loss at the unchanged locked route, measured against full-frame e15 in centred YCbCr444; the diagonal denotes equality. **b**, Exact-context loss versus an *ideal* shared-context synthesis MAC saving that charges each required feature cell once with zero overhead. Horizontal and vertical guides mark 0.1 dB and 10% saving. The latter is an analytical ceiling, not implemented sparse execution or wall-clock speed.

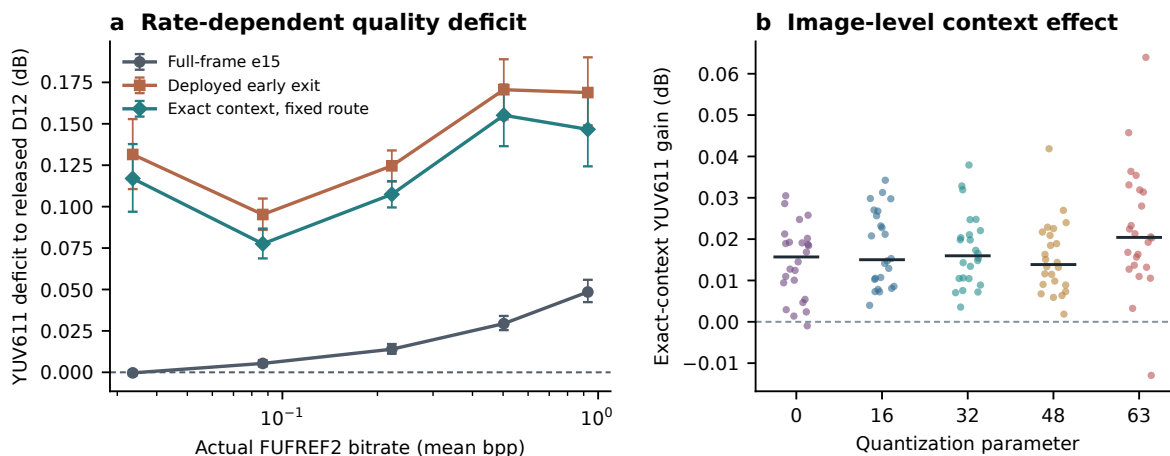


Figure S14. **Kodak24 exact-context transfer at unchanged real bitrates.** **a**, Mean YUV611 deficit to released D12 versus actual FUFREF2 bitrate for full-frame e15, diagnostic exact-context routing, and the deployed path. Vertical separations add to the released-to-deployed gap. Error bars are paired 24-image bootstrap 95% intervals; the released decoder is at zero deficit by definition. **b**, Per-image YUV611 gain from replacing zero-halo-plus-repair with clipped exact context and no repair, at each QP; dark bars mark medians. The exit maps remain fixed. No BD-rate or latency claim follows from this reconstruction-only counterfactual.

Sequence identity does not establish content independence. We subsequently screen all 1,378 first-frame pairs without using quality losses or router outputs. A fixed criterion combines a 63-bit low-frequency DCT fingerprint distance of at most six with centred 64×64 luma correlation of at least 0.98. Two pairs are flagged across folds: the 832- and 416-pixel-wide RaceHorses variants (distance 0, correlation 0.99267), and videoSRC17 with Kimono1 (2, 0.98039). Different raw hashes do not exclude related visual content. These are screening candidates, not proof of exhaus-

tive source or temporal identity. Conservatively joining them produces 51 proposed content groups for future splits. We do not change the existing folds or reselect their results after this audit; their scope remains sequence-disjoint development evidence, not content-independent generalisation.

We search a fixed router grid of 4,097 prices in $[-256, 256]$, which reaches both all-shallow and all-deep endpoints on every frame. Dither uses 385 mixture controls from zero to three; uniform uses the four reachable depths. Among controls satisfying a training-fold mean or

90th-percentile table-loss target, we maximise mean MAC saving. If none satisfies the target, we use the lowest training-risk control and explicitly retain its infeasibility. No test-source-conditioned fallback is applied. A held-out-label perturbation check confirms that test losses cannot change the fitted control.

At a 0.1 dB mean target, router and dither save 27.379% and 24.608% MACs, with mean held-out losses of 0.1006 and 0.0997 dB. Nevertheless, 122/265 and 123/265 cases exceed the target. Fitting the training-fold 90th percentile reduces these counts to 32 and 29, while savings become 17.586% and 15.558%. At the tighter 0.05 dB Q90 target, calibration groups covering 159/265 test cases are infeasible for each policy; these cases remain in every reported aggregate.

This distinction changes the deployment question: an average quality target, a bound on the frequency of large losses, and encoder-side verification of every frame are different contracts. Our table result supports investigating source-free control, but repair, cropping and reference alignment still require final-reconstruction replay before a delivered-quality claim. Section S16 performs this replay at the predeclared QP32; it also reports the unresolved paired routing margin.

S7. Separating the Histogram from Spatial Placement

A map can save arithmetic by selecting many cheap exits even if their locations contain no useful information. To isolate placement, let $p_k = N^{-1} \sum_t \mathbf{1}[k_t = k]$ be its fixed exit histogram and let π be a uniformly random permutation of its assignments. Each tile has marginal exit probability p_k , giving the exact expectation

$$\mathbb{E}_\pi \left[\frac{1}{N} \sum_t M_{t, k_{\pi(t)}} \right] = \frac{1}{N} \sum_t \sum_k p_k M_{t, k}. \quad (\text{S2})$$

We report $G = 10 \log_{10}(\mathbb{E}_\pi[\overline{M}_\pi]/\overline{M})$, where $\overline{M}$ is the recorded map’s table MSE. Positive G indicates useful spatial placement at exactly the same histogram and MAC cost. The logarithm is outside the expectation: this is not expected PSNR over random reconstructions.

At 0.1 dB, source-informed search has a mean placement gain of 0.0321 dB, the router 0.0213 dB, and dithering 0.0007 dB over their respective shuffled histograms. The router’s 95% sequence-cluster interval is [0.0167, 0.0262] dB; dithering’s is [−0.0002, 0.0017] dB. Uniform-depth maps have zero gain by construction. Among the 263 paired router maps, 204 have positive gain, seven have negative gain and 52 use one exit only. This diagnoses content association conditional on a source-calibrated histogram; it does not validate held-out autonomous calibration.

For each map we also solve the exact fixed-histogram assignment problem on M . Repeating each exit column

according to its count produces a square linear assignment problem. Its minimum table MSE gives a conditional upper bound on placement value. The router’s mean regret to that optimum is 0.0044 dB at the 0.1 dB target. Every source-search map attains its own fixed-histogram optimum, as expected for a separable Lagrangian minimiser. The optimum still requires the complete source table and carries the same information-acquisition cost. These exact table expectations are distinct from the following finite-permutation reconstruction intervention.

Actual reconstruction at unchanged depth proportions.

We replay all 53 QP32 frames using the fixed Q90 router and dither maps. Each original map is paired with three predetermined permutations, using seeds 202609280–202609282 and identical permutation indices for both policies on a frame. Assignments are shuffled only within strata of equal valid tile height and width. This preserves both padded depth counts and the valid-pixel area assigned to each depth. All 424 map cases are retained, including unchanged maps; caching identical maps requires 373 distinct reconstructions. Original-map MSE reproduces the earlier CPU replay.

For sequence i and policy p , we compute

$$G_{i,p} = 10 \log_{10} \left[\frac{\frac{1}{3} \sum_{s=1}^3 D_{i,p,s}^{\text{RGB}}}{D_{i,p,\text{original}}^{\text{RGB}}} \right], \quad (\text{S3})$$

then average sequences equally. Router and dither gains are 0.01393 dB [0.00949, 0.01930] and 0.00036 dB [−0.00031, 0.00105], respectively. Their paired difference is 0.01357 dB [0.00920, 0.01897]; resampling the 51 candidate-content groups gives [0.00923, 0.01883]. The intervals use 5,000 paired draws and condition on the three sampled permutations. They exclude permutation Monte Carlo uncertainty and training variation. This intervention measures placement within each policy’s own depth distribution. It neither matches the two policies’ achieved RD nor repairs the development-used calibration split. Table-optimal-map replay remains unevaluated; no bitstream or latency claim follows from these neural outputs.

S8. A Common Cap on Delivered YCbCr 4:4:4 Loss

The stored mixed reconstructions permit a second comparison under a common reporting anchor. For each frame–QP pair and policy, we collect its up to six archived nominal-budget maps, deduplicate identical maps, and retain only candidates whose measured cropped YCbCr 4:4:4 loss meets the chosen cap, with tolerance 10^{-4} dB. The evaluated full-frame e15 reconstruction is an explicit fallback with zero relative loss and zero MAC saving. The cheapest eligible candidate is selected; ties prefer lower loss.

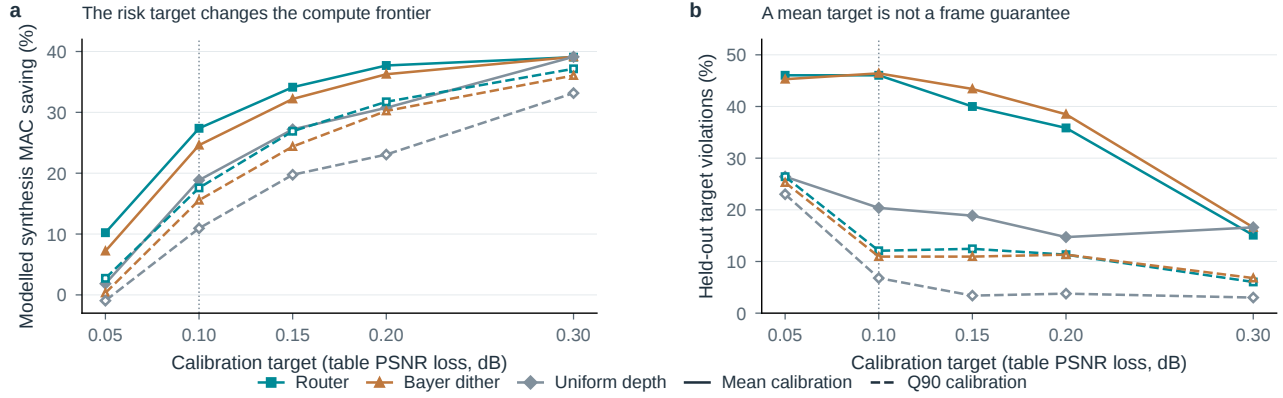


Figure S15. **A source-free control trades arithmetic savings against tail risk.** Controls are calibrated on other sequences, separately at each QP, using the training mean (solid) or 90th percentile (dashed). All 265 pairs are retained, including cases from infeasible calibration groups. **a**, Modelled MAC saving. **b**, Fraction exceeding the table-loss target. These padded source-error diagnostics do not establish final mixed-image quality, a per-frame guarantee or complete decoder speed.

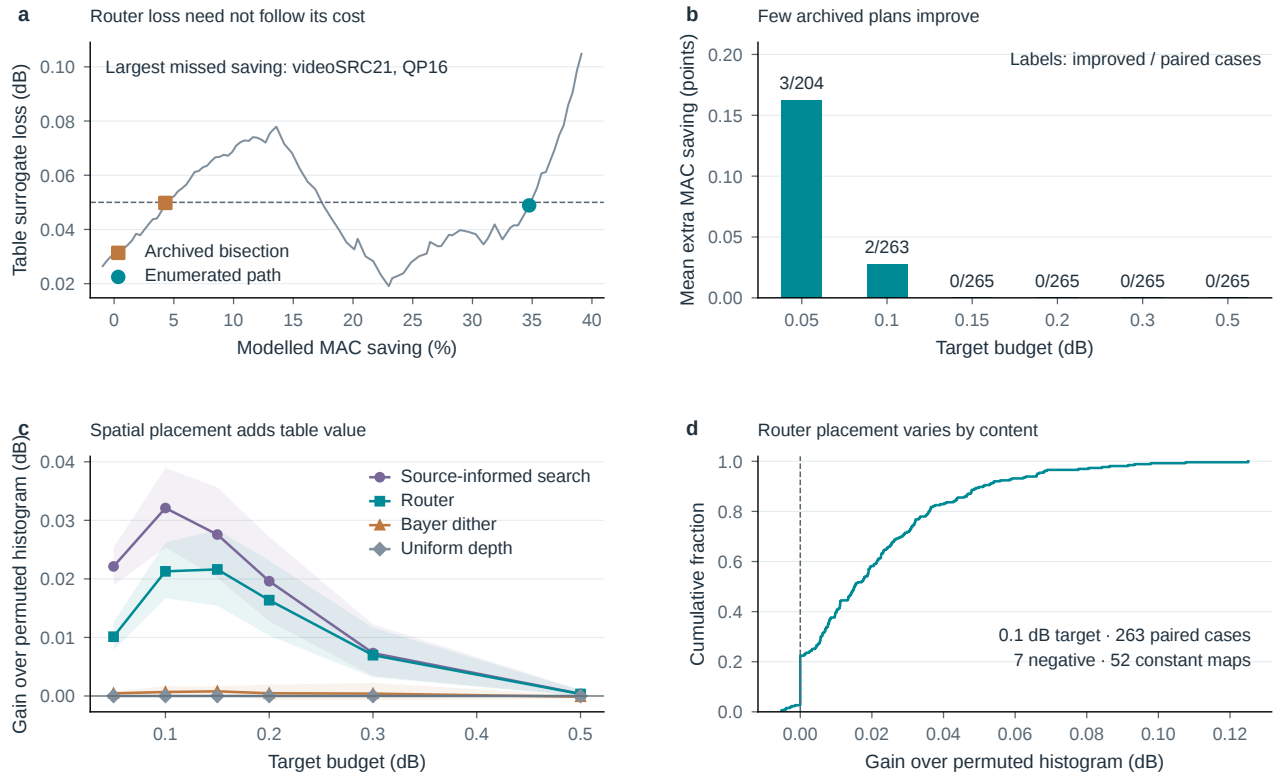


Figure S16. **Control-path and assignment diagnostics.** **a**, The explicitly selected largest missed router saving illustrates non-monotone table loss. **b**, Mean paired improvement from enumerating the router path; labels show improved and paired counts. Newly feasible cases are excluded from these means. **c**, Exact expected-shuffle comparison at unchanged exit histograms, with 95% sequence-cluster intervals. **d**, Router placement-gain distribution at 0.1 dB. All results are CPU analyses of padded error tables; none measures new final-image quality or latency.

This keeps all 265 pairs at every cap. At 0.1 dB, the retained maps save 29.19% of modelled synthesis MACs for source-informed search, 27.75% for the router, 24.81% for dithering and 22.98% for uniform depth. The router-dither difference is 2.93 percentage points, with a paired 95% sequence-cluster interval of [2.11, 3.79]. Their mean achieved YCbCr 4:4:4 losses are 0.0786 and 0.0791 dB, respectively. Full-frame fallback is selected for one router case, two search cases, two dither cases and eleven uniform cases. At a 0.3 dB cap, the routing margin falls to 0.35 points [0.11, 0.64] (Fig. S17).

Every selected output was measured in the original archive, and no new image or interpolated operating point is introduced. This diagnostic is nevertheless retrospective and source-aware: it selects among candidates after their final quality is known. It neither accounts for the cost of obtaining those candidates nor establishes an autonomous policy. Its finite candidate pools are not exhaustive, and its reference is fine-tuned e15, not the released model. Within those boundaries, the observed routing margin survives both a common delivered-loss cap and explicit fallbacks. The post-hoc 51-content-group bootstrap described above also retains the 0.1 dB margin: 2.9345 points with interval [2.1065, 3.7898], keeping all five QPs of every sampled sequence together. This checks sensitivity to the flagged content pairs; it does not remove source-aware candidate selection.

A descriptive influence check removes each sequence with all five QPs. At 0.1 dB, the remaining mean router-dither margin ranges from 2.80 to 3.12 points; this sensitivity range is not a confidence interval. Each QP also has a positive mean margin. Removing the three pairs where either method falls back leaves 2.91 points over 262 pairs, although this restricted subset changes the estimand. At 0.5 dB, 262 pairs tie and all three positive differences belong to one sequence. These checks are reproducible from `cap_influence_audit.json`.

S9. Shared-Exit Training Provenance

The released warm start, shared-exit fine-tuning and independent depth trainings are three distinct checkpoints/protocols. The archived shared-exit metadata lists 379,614 training images, batch size eight and 16 configured fine-tuning epochs. The inspected launcher selects 512-pixel crops through the late part of the upstream schedule, freezes the encoder/entropy path, and enables mixed-tile training, depth-scaled adapters, grid repair, adjacent feature distillation and a full-depth anchor. Its anchor and distillation coefficients are 10 and 1, respectively; the anchor code scales its batch-mean reconstruction error by the batch-mean λ unless the separate per-sample option is enabled. The e15 checkpoint stores weights, epoch and architectural configuration, not the complete launch command. Thus the launcher and metadata document the inspected intended setup, rather than proving

every historical process argument.

The evaluated router checkpoint explicitly stores live inputs `stem`, `qp`, soft-label cross-entropy, temperature ratio 0.1, training cost weight 1.3×10^{-5} , 48 stem projection channels and 256 hidden units. It has no spatial-context convolution. General-purpose latent/scale/bit pathways remain in the class, but do not provide variable content to this checkpoint. The decoder feature is pooled by tile mean and standard deviation, and the QP is divided by 63. Checkpoint inspection is recorded in `design_audit.json`. This corrects the broader description one would obtain by reading the router class defaults alone.

For tile errors $M_{t,k}$, exit costs c_k and $L_{t,k} = M_{t,k} + \lambda_c c_k$, the softened target is

$$p_{t,k}^* \propto \exp \left[-\frac{L_{t,k} - \min_j L_{t,j}}{\tau \max(\max_j L_{t,j} - \min_j L_{t,j}, \epsilon)} \right]. \quad (\text{S4})$$

The range normalisation controls the sharpness separately for each tile; $\lambda_c = 1.3 \times 10^{-5}$ and $\tau = 0.1$. This target is combined with the recorded regret term. It supervises a relative exit preference and does not calibrate the assembled image's loss.

The companion run record matches the checkpoint path, codec epoch, live inputs and stored evaluation statistics. It records 6,000 updates, batch size six, 512-pixel crops, seed zero, initial learning rate 10^{-3} and a regret coefficient of one. The inspected trainer uses AdamW with weight decay 10^{-4} , cosine scheduling and gradient clipping at norm one on Open Images. Its online agreement monitor withholds tiles within training images; the final label-agreement pass draws fresh images from the same shuffled epoch. Neither statistic is used as the manuscript's reconstruction or held-out calibration result. The stored head contains 148,176 parameters, including its inactive input branches.

S10. The Independent Depth Study

The ongoing experiment retains the first two, four or six synthesis trunk blocks of the DCVC-UF intra architecture and trains each complete codec under the pinned official recipe. The encoder and entropy weights learn alongside the decoder. Each model consequently has its own representation, distinct from the shared-latent exits evaluated in the main paper. The six-expert selector remains a proposed deployment path, not an implemented result.

Architectural parameter counts for all six depths are plotted in Fig. S27a. D2/D4/D6 have completed all 105 epochs in the final Kodak comparison below; released D12 has a different training history from the scratch runs.

The pinned training protocol. The source is Microsoft's image-training recipe at commit `cbdae87a5445` [4]. All 384,795 images in Open Images train_0/1/2 are retained as

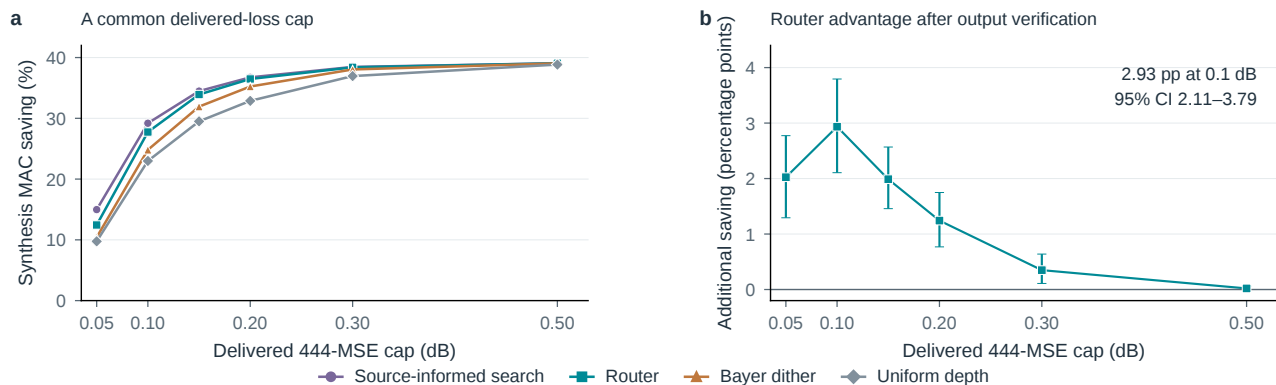


Figure S17. **Controlling delivered loss within the recorded candidate pool.** **a**, Maximum saving among actually measured eligible outputs, including a full-frame e15 fallback. **b**, Paired router–dither difference with 95% sequence-cluster intervals. All 265 pairs are retained, including explicit fallbacks, with YCbCr 4:4:4-loss tolerance 10^{-4} dB. This retrospective source-aware comparison excludes the cost of candidate generation and makes no released-reference or latency claim.

lossless decoded-RGB PNGs, with pixel equality checked during preparation. Each independent model uses seed 42, Xavier initialisation, batch size 16, AdamW with the upstream defaults, and gradient clipping at norm 0.1. QP is sampled from 0–63 with logarithmic λ interpolation from 10 to 2048. The original loss, including its literal 0.0833 coefficient, and the upstream crop, flip and colour transforms are retained. Figure S18 records the full 105-epoch schedule. There are 24,049 batches per epoch and 2,525,145 scheduled updates per model. With the official shuffled `drop_last=True`, eleven images per epoch do not enter a batch; no image is permanently excluded from the training manifest.

Training uses FP32 with the upstream TF32 flags and `torch.compile`, with `fallback_random=True` to retain the PyTorch random-number path. AMP is disabled. The instrumented eager step was checked against the upstream step at identical inputs and RNG state. Compilation can change floating-point operation order, so protocol parity does not imply bitwise-identical training trajectories. Check-point manifests retain the execution mode and source hashes.

Finished D2/D4/D6 at matched estimated rate. All three completed models were evaluated on the same 24 full-resolution Kodak images and five QPs. For each image, monotone PCHIP interpolation in log *estimated* bpp aligns quality without extrapolation. At 0.2 bpp, D4 improves on D2 by 0.116 dB RGB PSNR [0.094, 0.139], while D6 improves on D4 by 0.064 dB [0.052, 0.077]. The corresponding YUV 6:1:1 gains are 0.120 dB [0.096, 0.143] and 0.058 dB [0.045, 0.071]. Against the released D12, image-mean YUV BD-rate costs over the common four-model PSNR interval are +7.99% [7.11, 8.82] for D2, +4.54% [4.06, 5.01] for

D4, and +2.79% [2.43, 3.15] for D6. Intervals use 5,000 paired Kodak-image bootstrap draws and exclude training-seed variability. Figure S19 depicts this capacity trend. Rates are deterministic entropy estimates, not emitted bitstream lengths; released D12 is a historical reference rather than the matched scratch D12 control. The comparison cannot attribute the entire gap to depth.

Synthesis versus complete neural-decoder arithmetic.

We trace every convolution in the pinned architecture, counting each use of the causal spatial prior. At 512×512 , the unchanged neural entropy-recovery path costs 31.079 GMac. D12 synthesis contributes 56.915 GMac, while D2 contributes 14.495 GMac. Thus a 74.53% synthesis reduction becomes a 48.21% reduction when those fixed recovery networks are included (Fig. S20). These are architectural counts, not trained-quality or timing results. Meta-tensor traces were checked against a real CPU D2 forward at 64×64 ; the repeated prior is counted three times. Biases, elementwise operations, rANS, memory movement and routing remain outside this Conv2d accounting.

The native encoder also produces a reconstruction while its CPU entropy worker can run concurrently. Even including synthesis in an encoder MAC total therefore does not justify adding isolated stage times or converting MAC reduction into wall-time reduction. The same execution contract and precision must be used for every depth in a future paired timing study.

Isolating depth. Released D12 is a useful deployment anchor, but a D12 trained with the same data manifest, 105-epoch schedule, optimisation and initialisation protocol is necessary to isolate depth from training history. D8 and

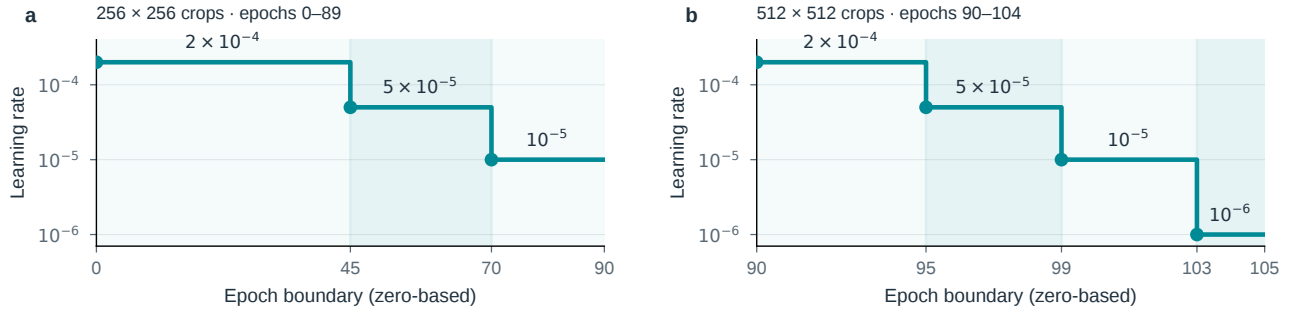


Figure S18. **The full official schedule is retained.** **a**, First 90 epochs with 256^2 crops. **b**, Final 15 epochs with 512^2 crops, on a separately expanded epoch axis. Zero-based boundaries delimit the seven learning-rate stages; the protocol applies to D2/D4/D6. These are prescribed settings, not learning curves.

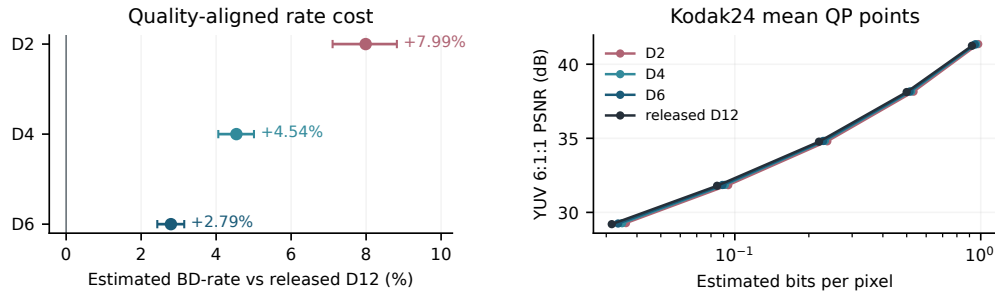


Figure S19. **The completed independent codecs approach the release as depth grows.** Left, per-image YUV 6:1:1 BD-rate cost on the four-model common PSNR interval; bars are 95% paired-image bootstrap intervals ($n = 24$). Right, mean five-QP operating points on Kodak24. Rates are deterministic entropy estimates, not actual bitstreams. Released D12 differs in training history from the three scratch models.

D10 should be prioritised after the initial depth frontier and this control are available. The main runs retain the official recipe; distillation and prefix initialisation are separate variants rather than silent training changes.

Isolating routing. The evaluation should progress from full-frame D12 to patchwise D12, the best validation-selected fixed expert, blind mixtures, simple image statistics, the MLP and source-informed search. Full-frame versus patchwise D12 measures the cost of tiling and entropy resets before routing receives credit for any gain. A histogram-preserving shuffle separates the chosen expert proportions from their spatial placement. A model below the dataset-mean frontier may still help a subset of patches. Expert removal should therefore be evaluated by leave-one-expert-out routing at matched constraints, not justified by mean full-image quality alone.

A concrete router hypothesis. A lightweight encoder-side predictor can estimate each expert's distortion and rate relative to a cheap reference. Let $z_{t,k}$ indicate the expert assigned to patch t and $n_k = \sum_t z_{t,k}$. For equal-shaped patches and measured grouped-runtime tables $\tau_k(n_k)$, a proposed system objective is

$$\hat{J} = \sum_{t,k} z_{t,k} (\hat{D}_{t,k} + \lambda \hat{R}_{t,k}) + \mu \sum_k \tau_k(n_k). \quad (S5)$$

Routing, packing and scatter/gather work must also enter the measured total; the expression isolates the expert-runtime dependence. An independent per-patch latency label approximates this coupled problem. A bounded post-processing variant could move a patch from k to j according to its predicted RD change and the measured marginal time $\tau_k(n_k - 1) - \tau_k(n_k) + \tau_j(n_j + 1) - \tau_j(n_j)$. The number of passes and decision-time limit should be fixed on validation images. Any improvement must include this extra control work.

Patch counts alone cease to describe the workload when

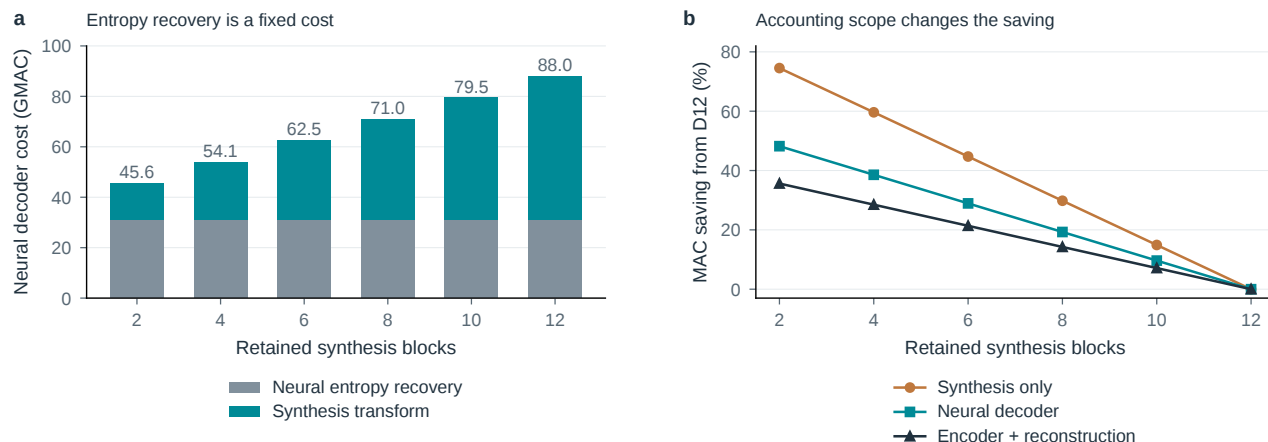


Figure S20. **The arithmetic denominator changes the apparent saving.** **a**, Synthesis varies with depth while neural entropy recovery stays fixed. **b**, The same architectural intervention under three accounting scopes. Counts use 512×512 input and one MAC per multiply-accumulate; they are not observed kernel times. D8/D10 are architecture-only controls. Encoder totals include reconstruction but omit rANS, elementwise work, transfers and control overhead.

neighbouring regions are coalesced or frame edges create smaller windows. Then the assignment induces a set of coded regions, each with its own shape and context. Runtime tables must condition on expert, shape, QP and the validated execution mode; packing regions of different shapes into a hypothetical common batch is not an implemented optimisation. Region construction also changes repeated context, entropy resets and headers. Accordingly, the fixed-map merge control separates these execution choices from prediction accuracy. A same-histogram spatial shuffle is a useful additional control: equal expert frequencies need not imply equal grouping opportunities or equal codec cost.

Regression or regret-weighted training should be compared against ordinary exit-label cross-entropy. A second-stage top-two verification can be restricted to small predicted margins, with its threshold selected on held-out calibration images and its extra encoding work included. Neither a large prediction margin nor a low average regression error establishes a framewise quality guarantee. These are prospective experiments, not measured components of the current shared-exit system.

Spatial competition between complete codecs already provides a close precedent [1]; adaptive compression and patch early exit provide further controls [2, 5, 7]. The testable direction here is depth-heterogeneous selection with measured rate-distortion-time costs and reduced encoder search. Practical codec studies motivate including device execution and perceptual assessment [3, 6].

S11. Evaluation Contract for the Next Stage

Expert training, router training, calibration and final evaluation should be split at image level. Patches from the same

image must stay in one split. The four fixed validation crops in the current training monitor are health checks; a broader intermediate evaluation should use the available 100 DIV2K validation images, followed by the declared final Kodak/CTC/CLIC scope. The frozen epoch-20 DIV2K crop evaluation below is complete; the final trained-model, full-image and native-runtime study remains pending.

Each codec needs a true-byte rate-distortion curve. Equal QP does not imply equal rate across independently learned codecs. Comparisons should use their common measured support, reporting infeasible targets and avoiding extrapolation. Rate includes main and hyperprior payloads, model identity, quality/control parameters and framing headers, divided by original pixels. Quality is measured after cropping valid pixels; the exact RGB/YUV formula and perceptual metrics are declared before test-set selection.

Timing requires the same GPU, precision, kernel path and warmup, with alternated paired execution order and retained raw trials. Encoder, decoder and complete round-trip times are separate outcomes. CUDA-event module time and end-to-end wall time must remain separately labelled. Grouping, scatter/gather, entropy work, stream parsing, transfers and fallback are included where they occur. The full codec bank's resident weight memory and peak activation memory are both reported.

The primary success criterion is a positive net runtime advantage over the best validation-selected blind control on common achieved rate-distortion support. Critical comparisons should then be repeated across training seeds. A content bootstrap at fixed weights measures a different source of variation and cannot replace those repeats.

S12. Frozen Interim Depth Models: Actual Coded Rate

The four-crop health monitor cannot establish a rate-distortion frontier. We therefore freeze the D2, D4 and D6 checkpoints after exactly 20 of the 105 scheduled epochs and evaluate all 100 DIV2K validation images as 512×512 centre crops. The released D12 checkpoint is a practical anchor with different training provenance, not the control that isolates the effect of depth. All four models are evaluated at QPs 0, 16, 32, 48 and 63: 2,000 encode/decode cases. No checkpoint is selected using these results. The study is an interim diagnostic, not the final depth ablation. We repeat the complete protocol at the fixed epoch-30 milestone below; the plotted geometry and allocation controls retain their epoch-20 weights.

A causal CPU reference. Each stream contains actual rANS payload. An encoder-independent process reconstructs the image using the stream and resident model, with source analysis disabled. It first decodes the hyperlatent and then the four causal main-latent stages; entropy indexes come from the decoder's recovered context, rather than being supplied by the encoder. Every case has exact reconstruction, latent and symbol/index-trace checks. The decoder process is reused within a model, so independence does not imply a newly launched process for every image. The reference uses CPU FP32 and a versioned research format, not the released CUDA FP16 format or a latency benchmark.

We distinguish entropy estimates, rANS payload and container bytes. The research header is 88 bytes per crop, including 64 bytes of identity and integrity hashes: 0.002686 bpp on a 512^2 crop. This is not a minimal production header or the cost of a routing map. Reconstructions are explicitly converted from clipped YCbCr to clipped RGB before quality measurement. These RGB results are separate from the legacy shared-exit archive's equal-channel YCbCr-MSE metric.

Compare rate, not quality index. For each image, we interpolate PSNR linearly in log payload rate and retain only the common support of all four codecs. No quality is extrapolated. The supported cohorts contain 95, 99 and 89 images at 0.1, 0.2 and 0.4 bpp respectively. At 0.2 bpp, D4 and D6 improve over D2 by 0.1207 dB [0.1018, 0.1420] and 0.1834 dB [0.1572, 0.2115]; the released D12 gap is 0.6986 dB [0.5989, 0.8112]. Intervals use 5,000 paired-image bootstrap draws, conditional on the fixed checkpoints and included images. They do not measure training-seed variation. PCHIP interpolation gives 0.1209, 0.1839 and 0.6880 dB on the same 99-image support. Figure S21 separates native-QP population means from these paired, equal-rate comparisons.

Why the small monitor was insufficient. The original four crops and the additional 96 are reported separately on the same supported cohort (Fig. S22). The four-crop subset overstates the mean depth advantage at 0.2 bpp; it is useful for training health, not a substitute for the validation distribution. D6 nevertheless retains an approximately 0.18 dB mean advantage over D2 across the larger cohort. Its modest magnitude is therefore not solely an artefact of monitoring four images. The remaining released-model gap cannot be attributed to depth until an official-recipe D12 control has been trained for the same schedule.

BD-rate is computed per image using PCHIP interpolation of log rate as a function of PSNR, over the intersection of all four models' PSNR ranges. All 100 images have a valid interval. Relative to released D12, mean payload BD-rate is +21.56% for D2, +17.84% for D4 and +15.69% for D6 (Fig. S23). This compares unfinished shallow models with a released checkpoint; it is not the bitrate penalty caused by truncation at matched convergence. Averaging per-image BD-rate and comparing population-mean RD curves are different estimands.

Milestone sensitivity. The next fixed checkpoint at epoch 30 repeats all 2,000 encode/decode cases. The crop manifest, CPU backend and research format are unchanged; all 500 released-D12 measurements reproduce exactly. To compare depth ordering, Fig. S24 retains the same 99 images on common 0.2-payload-bpp support across both milestones and all four models. D6's mean advantage over D4 changes from 0.0627 dB to -0.0045 dB, whose interval includes zero. Relative to their own epoch-20 curves, D2 and D4 gain 0.0259 and 0.0168 dB while D6 loses 0.0504 dB on this support. These are descriptive checkpoint differences, not evidence that training a deeper model is intrinsically harmful. Both milestones are retained without quality-based checkpoint selection; neither establishes the final ordering after 105 epochs.

S13. Does an Additional Expert Add an Additional Choice?

A bank with more models has greater capacity, but the value of its middle experts depends on the attainable allocation frontier. We perform a conditional leave-D4-out analysis on the measured epoch-20 D2/D4/D6 curves. The decision unit is a whole 512^2 crop, not a patch within an image. All three choices are compared at the same interpolated payload rate for each image. Dynamic programming minimises summed RGB MSE under an aggregate neural-decoder Conv2d MAC budget, permitting unused budget. The objective is reported as PSNR of pooled MSE, not mean per-image PSNR.

Figure S25 separates two questions. Removing D4 measures its incremental option value relative to the best

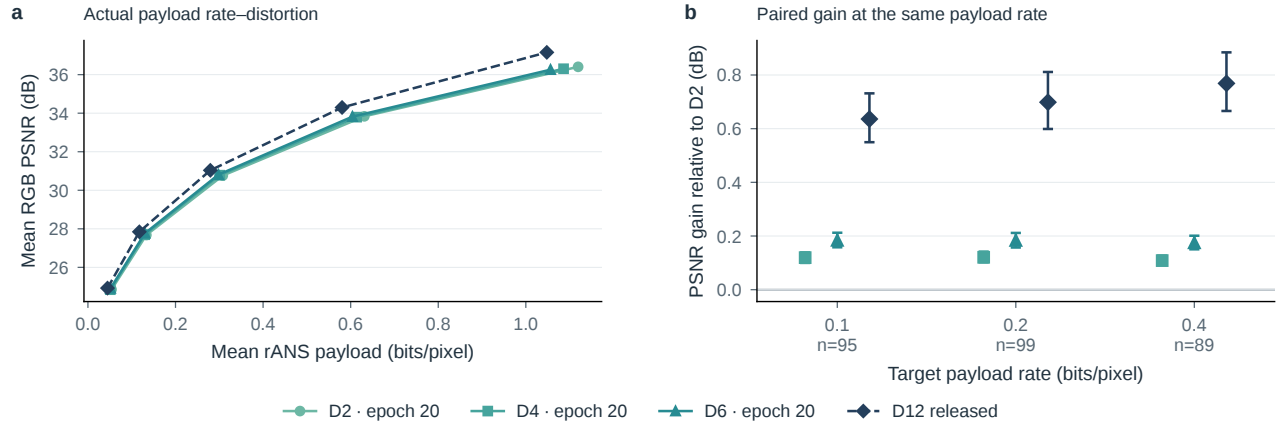


Figure S21. **Shallow models are close, but not identical, at equal coded rate.** **a**, Mean RGB PSNR and actual payload over all 100 crops at five QPs; joining lines are visual guides. **b**, Per-image interpolation and paired differences on four-model common support, with 95% image-bootstrap intervals. D2/D4/D6 are frozen epoch-20 models; released D12 is not matched in training history. These measured streams establish an interim CPU rate–distortion comparison, not final quality or native runtime.

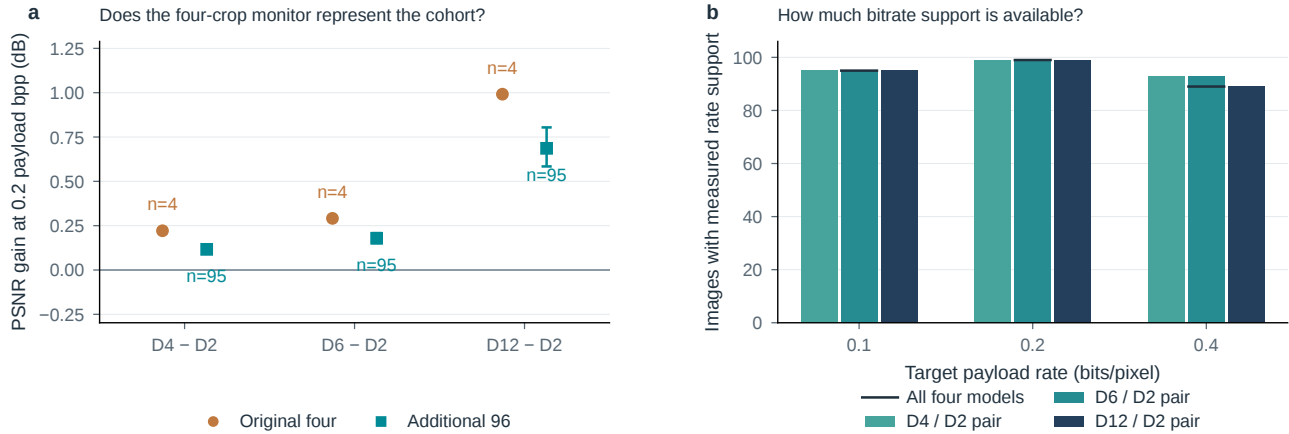


Figure S22. **Coverage is part of the result.** **a**, Original monitor crops and additional images at 0.2 payload bpp, restricted to four-model common support. We do not estimate uncertainty from the four-crop subset. **b**, Pairwise and four-model supported counts at each target rate. A changing cohort is made explicit rather than filled by extrapolation.

D2/D6 assignment. Permuting the labels of the best source-calibrated blind histogram measures the value of knowing where to place depth. The blind comparator minimises expected MSE on this same cohort; it is deliberately optimistic and is not a deployable validation-fitted policy. At a budget equal to uniform D4, adding D4 is worth 0.0132, 0.0086 and 0.0061 dB at 0.1, 0.2 and 0.4 bpp. Source-informed placement over the blind expectation is worth 0.0239, 0.0303 and 0.0299 dB. These are deterministic, conditional bounds at one interim checkpoint. They motivate expert-removal controls, rather than establishing that a six-expert spatial bank is necessary.

Complexity is not the training target. We also report every combination of three cheap source statistics, two depth comparisons and three rates: RGB standard deviation, gradient energy and Laplacian energy versus D4–D2 or D6–D2 benefit (Fig. S26). At 0.2 bpp, gradient energy has Spearman correlation 0.270 with D4–D2 but only 0.111 with D6–D2; Laplacian energy has correlation 0.011 with D6–D2. This does not rule out a learned cheap representation. It does caution against treating visible texture as a sufficient label for incremental reconstruction benefit. The statistics are descriptive associations on development-used data, without significance selection or held-out prediction claims. A later selector must be trained and assessed on separate images using rate, distortion and measured cost.

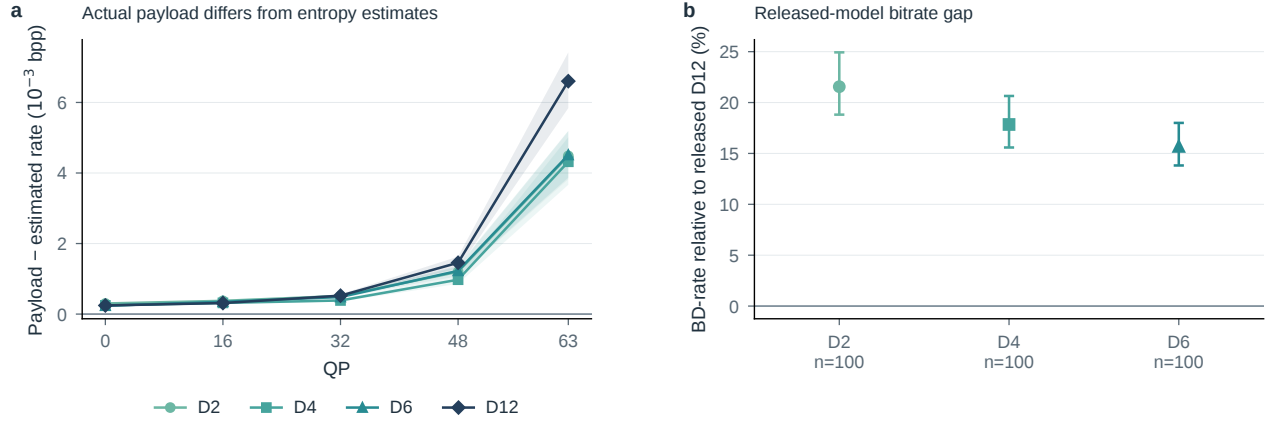


Figure S23. **Estimated entropy and emitted bytes are separate quantities.** **a**, Actual rANS payload minus entropy estimate, excluding the research header. **b**, Mean per-image BD-rate against the released anchor on each image’s four-model common PSNR interval. Intervals resample images; no uncertainty from independent training seeds is represented.

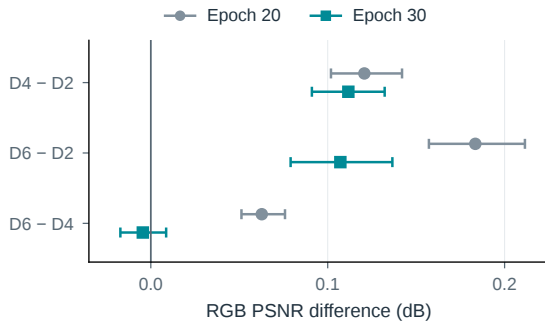


Figure S24. **Intermediate depth ordering changes during training.** Paired RGB PSNR differences at 0.2 actual payload bpp on the same 99 images. Epochs 20 and 30 are retained without quality-based selection. Intervals are 95% image-bootstrap intervals at fixed weights, excluding training-seed uncertainty. No rate extrapolation.

We sweep $f \in \{0.5, 0.7, 0.9\}$ and $h \in \{0, 0.03\}$. These are explicit sensitivity assumptions, not fractions estimated from the hardware. The reciprocal of Eq. (S6) is the plotted scenario speedup. It shows why removing most trunk blocks need not yield a proportional reduction in complete codec time. The curves deliberately contain no predicted PSNR, bitrate or empirical uncertainty bands.

Exact decoder size is 1,767,168 + 1,038,720d parameters; adding the 27,947,520 non-synthesis parameters gives complete-codec size. This formula matches all three independent training manifests. It describes capacity, not compute: retained tensors, arithmetic and measured device time are separate quantities. The data file labels these scenarios explicitly and records every assumption used to generate them. A separate CPU construction of all six depths verifies these counts and exact equality of their retained initial tensors to the seed-matched D12 initialisation; it performs no training or quality evaluation.

S15. Encoder Information Acquisition

Figure S28 separates candidate reconstruction from error-table construction, control search and the final mixed reconstruction. A single trunk pass can tap all reachable depths, reducing repeated prefix work. The historical timing summary lacks raw repetition uncertainty; it cannot be used as a paired end-to-end benchmark. In addition, the archived GPU search timer may include pending reference work, so it is excluded from the isolated search comparison. The runtime audit in the paper repository also identifies an older neural replay benchmark whose outputs are not causally reconstructed from decoded stream symbols. Its stage labels must not be read as a verified complete codec.

S14. Analytical Scenarios for the Pending Runtime Measurements

The independent depth study has an interim CPU RD diagnostic, but no final trained-model RD or measured native-latency frontier yet. To make the design hypothesis inspectable, Fig. S27 separates exact architectural parameter counts from a hypothetical runtime model. For retained depth d , let f be the assumed fraction of full-depth time spent in the twelve-block trunk and let h be added routing/grouping overhead normalised by the original full-depth time. Under linear trunk-time scaling,

$$\frac{T(d)}{T(12)} = (1 - f) + f \frac{d}{12} + h. \quad (\text{S6})$$

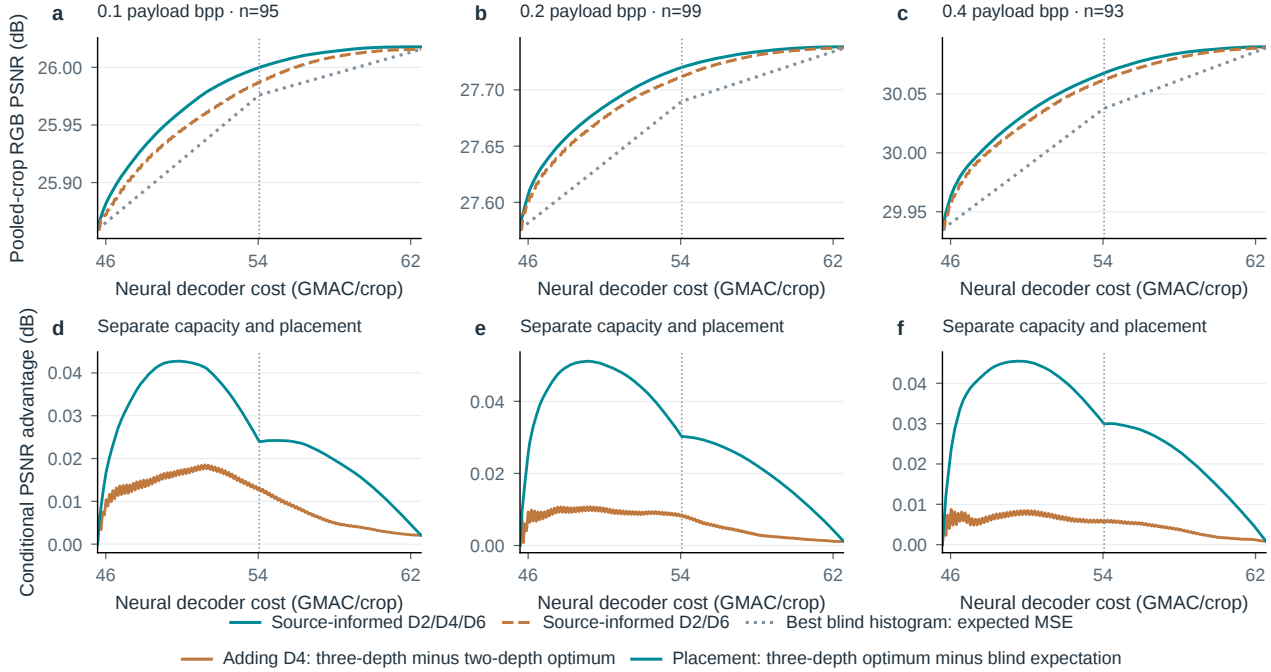


Figure S25. **Separate the value of capacity choices from their assignment.** Top: exact whole-crop allocation under a shared neural-decoder MAC budget. Bottom: incremental D4 option value and assignment premium over the best source-calibrated blind histogram’s expected MSE. Vertical guides mark the uniform-D4 budget. The curves use interpolated equal-rate quality and contain neither patch entropy resets nor a trained MLP. They are not within-image routing results or runtime frontiers.

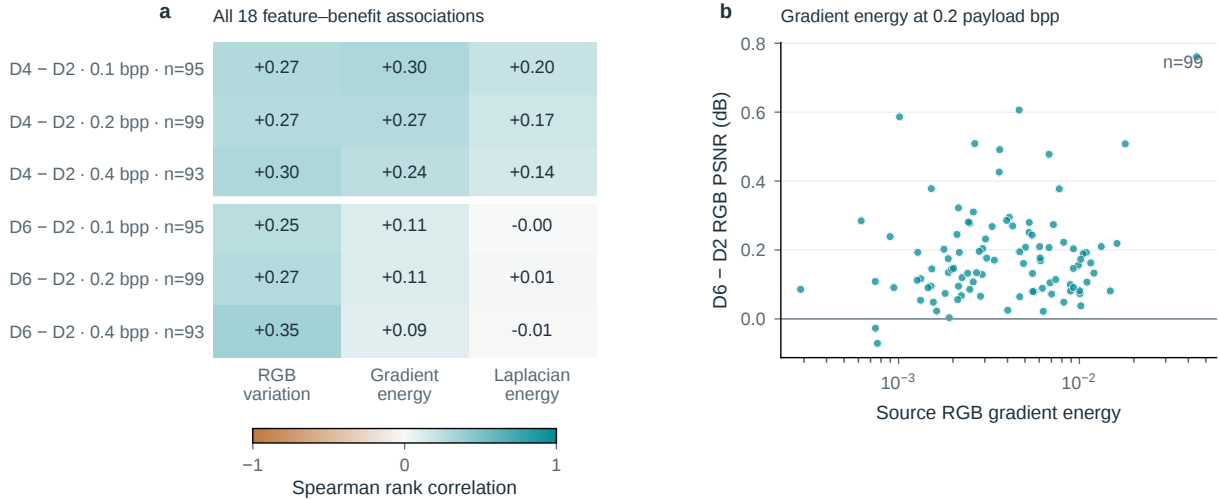


Figure S26. **Cheap complexity measures do not directly identify the useful expert.** All 18 specified feature/depth/rate combinations are shown, with supported image counts. The scatter panel was fixed to gradient energy versus D6–D2 at 0.2 bpp before inspecting the associations. These are correlations with interim equal-rate benefits, not router accuracy or causal feature effects.

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S16. Fixed Controls on Actual Reconstructions

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A stopping policy must eventually be judged on its reconstructed image, rather than only on the table used to choose

its control. We therefore replay the sequence-disjoint controls of Sec. S6 at the predeclared QP32 and nominal 0.1 dB target. Every one of the 53 first frames is evaluated with mean and Q90 calibration under router, dither and uniform

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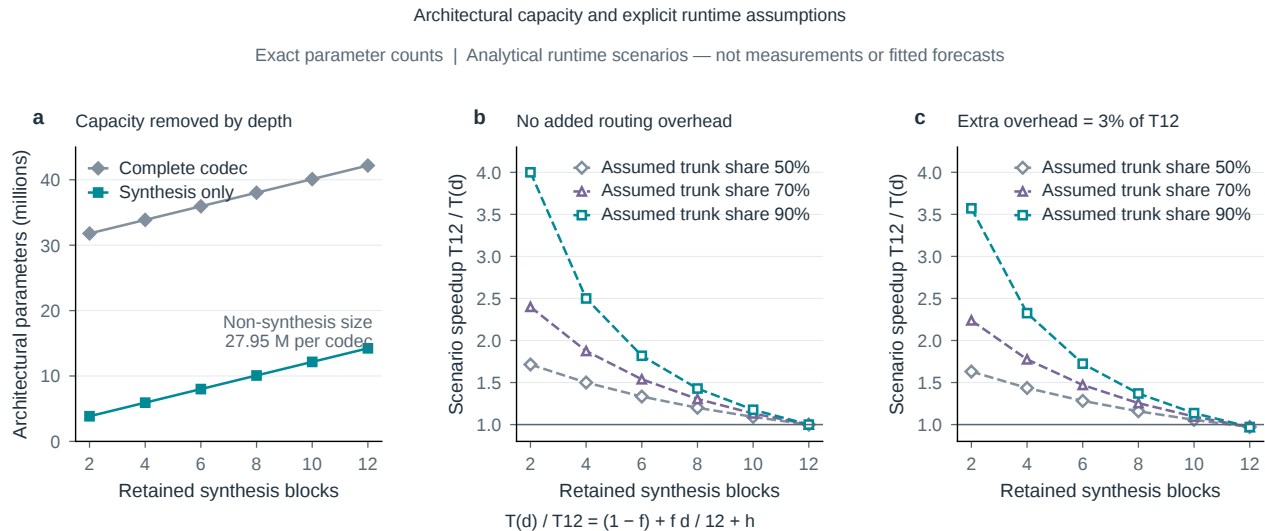
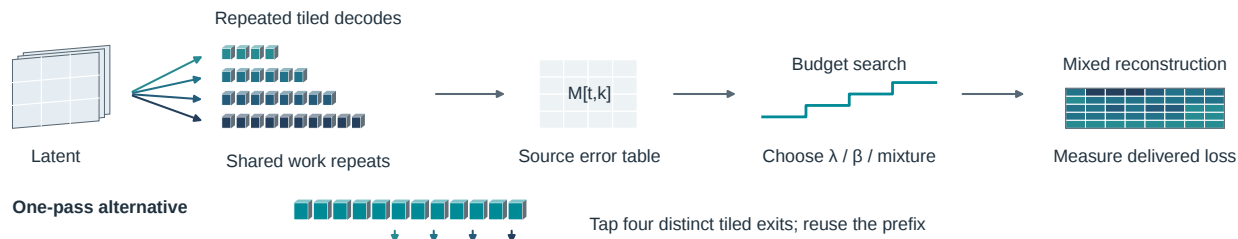
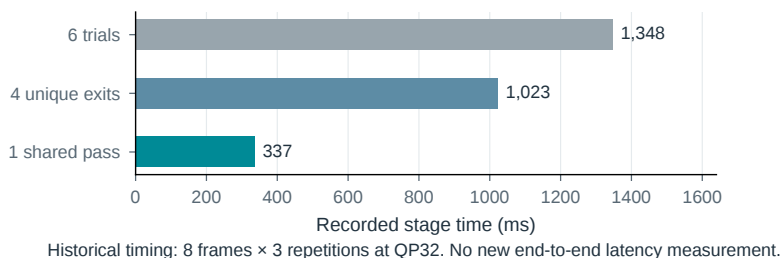


Figure S27. **A transparent design-space illustration for unfinished experiments.** **a**, Exact architectural parameter counts. **b,c**, Analytical runtime sensitivity to assumed trunk-time share and added overhead. Open markers and dashed lines denote assumptions, not measurements or fitted forecasts. No quality or bitrate outcome is assigned to an unfinished model.

a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



c Calibration changes the cost

Fixed validation-selected β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t,k]$.
A fast router does not remove that cost.

Figure S28. **The decision path has an encoder cost.** **a**, Stages required for source-error-based selection, with prefix reuse across candidate exits. **b**, Historical table-construction upper medians over 24 pooled trials, without invented uncertainty bars. **c**, A fixed held-out control and per-frame source calibration require different information. This is a stage analysis, not an end-to-end speedup.

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policies: 318 actual neural reconstructions. The control and exit map are fixed before the source image is opened for reconstruction and scoring. No held-out source-error table or quality-based fallback enters this decision. The checkpoint, sequence folds and development-used corpus remain fixed;

this is not an independent test of trained-router generalisation.

The final reference is the same ϵ_{15} full-frame reconstruction for every policy. We report cropped, unclipped equal-channel YCbCr loss and separately convert and clip

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1048 RGB before computing its MSE-ratio loss. The RGB
1049 source is obtained from the archived YCbCr420 input; it
1050 is not the original camera RGB. Against this common refer-
1051 ence, mean-calibrated router and dither save 25.675% and
1052 23.802% synthesis MACs, at mean RGB losses of 0.08679
1053 and 0.08814 dB. Q90 calibration changes these values to
1054 16.741%/14.204% and 0.05684/0.05636 dB. The correspond-
1055 ing RGB target-exceedance counts are 17/16 under mean
1056 control and 4/2 under Q90, each out of 53 images (Fig. S29).
1057 A mean or quantile calibration target is therefore not a per-
1058 image guarantee.

1059 **A positive average is not a resolved routing advantage.**
1060 The paired router-minus-dither saving is 1.87 percentage
1061 points under mean control, with a 95% sequence-bootstrap
1062 interval of $[-1.60, 5.19]$. Under Q90 it is 2.54 points, with
1063 interval $[-0.42, 5.54]$. Both include zero. Moreover, equal
1064 control targets do not enforce equal achieved RGB quality.
1065 We retain all sequences and report these intervals; this replay
1066 does not establish a positive equal-quality routing premium.
1067 It complements, rather than replaces, the per-source retro-
1068 spective delivered-cap comparison in the main paper.

1069 As a post-hoc sensitivity check, we resample the 51 con-
1070 servative content groups identified in Sec. S6, keeping all
1071 members of a sampled group together and retaining equal
1072 sequence weights. With 5,000 draws, the corresponding
1073 saving intervals are $[-1.91, 5.52]$ and $[-0.64, 5.62]$ points;
1074 both still include zero. Point estimates and policies are un-
1075 changed. This sensitivity analysis does not repair the original
1076 fold assignment or prove that the proposed groups exhaust
1077 all related content.

1078 On identical padded support and against the archived
1079 scalar reference, the largest absolute discrepancy between ac-
1080 tual replay and the stored table prediction is 3.96×10^{-4} dB
1081 across all 318 cases. This is a numerical consistency ob-
1082 servation. It does not separate backend differences from
1083 mixed-head interactions, nor validate entropy-stream decod-
1084 ability. The experiment reconstructs neural outputs on CPU;
1085 it is neither a native GPU bitstream test nor a latency mea-
1086 surement.

1087 **A bounded stream-only decoder check.** A separate en-
1088 gineering preflight uses the first BasketballPass and BQ-
1089 Mall frames at QP32, fixed before execution. Each uses its
1090 archived mean/Q90 router map, an all-deep map and a cyclic
1091 6/8/10/12 map: eight cases. A versioned CPU research
1092 container carries the valid dimensions, e15 checkpoint iden-
1093 tity and two-bit exit indices around the common FUFREF1
1094 entropy stream. A separate decoder process accepts only
1095 stream/output paths; source analysis and hyperanalysis calls
1096 are disabled. All eight cropped reconstructions, recovered
1097 latents, symbols and entropy indexes match encoder-side

reconstruction exactly. Seven malformed-stream checks also
pass.

For each frame, all four maps retain identical inner en-
tropy bytes. BasketballPass and BQMall carry 2,198 and
10,954 entropy-payload bytes, plus 1 and 2 map bytes, re-
spectively; each adds 176 bytes of explicit research headers.
These headers are not a minimum production format. This
small CPU check establishes neither native CUDA interop-
erability nor 53-frame bitstream coverage, autonomous map
selection or deployment speed.

Recompute the predictor, not only its saved decisions.
A second CPU audit reloads the frozen codec and router
checkpoints and recomputes source analysis, shared-stem
features and router logits on all 53 frames. Historical GPU
and current CPU log-probabilities are not exactly equal: their
largest absolute difference is 0.02266. Nevertheless, both
fixed cross-fit controls reproduce every exit decision: zero
changed tiles out of 1,765 under either mean or Q90 control.
This validates the replayed maps at these particular settings,
not every control value, QP or backend. Forward hooks
count 289.715 router Conv2d/Linear MACs per padded im-
age pixel. Pooling, normalisation, nonlinearities, selection,
memory movement and dispatch are excluded; this arith-
metic count cannot establish router latency.

S16.1. The released decoder is a separate quality anchor

We now close the reference gap on the same 53 QP32 frames.
A strictly loaded stock DMCI decoder uses the official re-
leased D12 weights. Before reusing the latent, we verify
exact equality of all 255 analysis, entropy and quality scale
tensors with e15. The first frame additionally requires exact
agreement between stock synthesis and its remapped warm-
start implementation. Input padding and valid-pixel support
are identical to the shared-exit replay: replicate to multiples
of 256, then crop before scoring. This is CPU FP32 neural
reconstruction, not the native image-pad16 CUDA bitstream
path.

Released D12 reaches a mean RGB PSNR of
34.71763 dB, compared with 34.70855 dB for e15 full-frame
synthesis. The paired e15 loss is 0.00908 dB, with a 95%
sequence-bootstrap interval of $[0.00785, 0.01033]$ dB. We
then express all 318 fixed-policy outputs against this released
reference without changing their maps, reconstruction errors
or costs (Fig. S30). Mean-control router and dither lose
0.09587 and 0.09722 dB RGB PSNR relative to released
D12; their Q90 losses are 0.06592 and 0.06544 dB. The
counts above 0.1 dB become 20/20 under mean control and
7/5 under Q90, each out of 53.

Changing the common reference adds the same per-image
offset to every policy's loss. It changes absolute quality
and threshold counts, but cannot change the paired router-

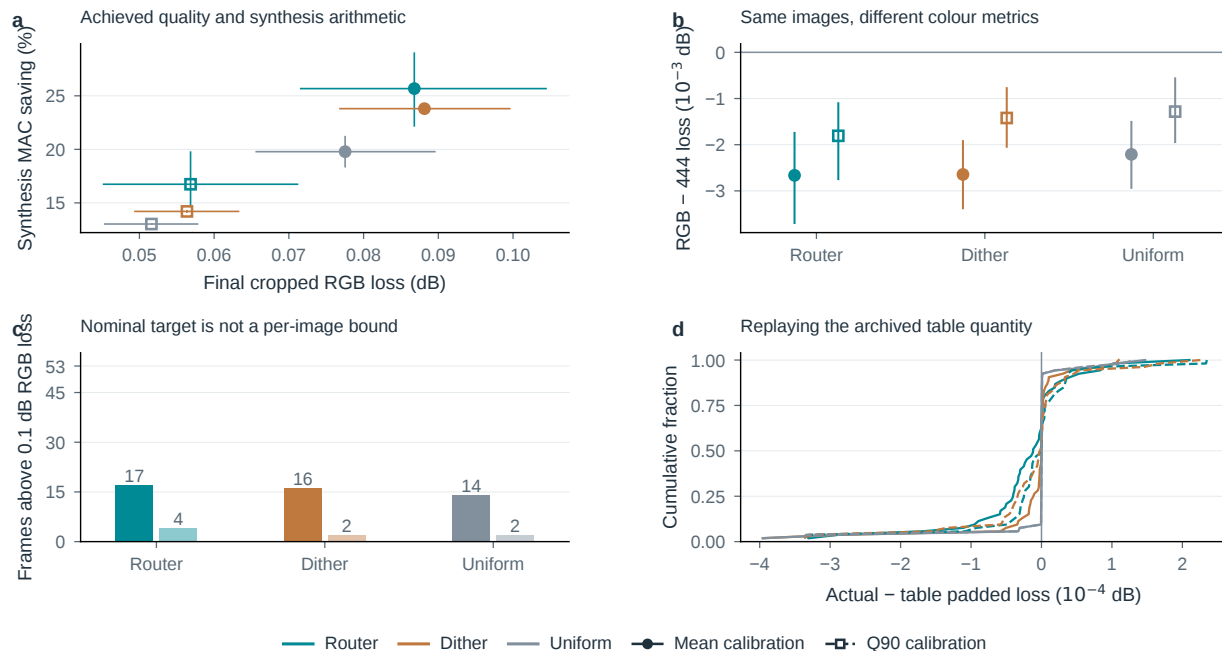


Figure S29. **Carry the fixed decision through to the actual image.** All 53 sequences at QP32, with no outcome-based exclusions. **a**, Achieved RGB loss and synthesis MAC saving with conditional 95% sequence-bootstrap intervals; these points are not an equal-quality frontier. **b**, RGB-minus-444 loss on the same support and reference. **c**, Counts above the nominal numerical threshold; pale bars denote Q90 calibration. **d**, Actual-minus-table loss on padded support. Mean and Q90 controls use a different calibration reference and metric from the cropped RGB report. The paired router-dither saving intervals include zero under both controls.

dither loss or MAC difference. The original controls were calibrated on padded 444 errors, so these cropped RGB counts are a reference-alignment diagnostic, not a failed RGB guarantee that the policy was trained to provide. The result closes this particular QP32 comparison; it does not replace the five-QP archive with a new metric or establish native deployed quality and latency.

S16.2. Which learned components carry the reconstruction?

A component can look dispensable in an architecture diagram yet carry a substantial learned correction. We hold the e15 weights and all 53 Q90 router maps fixed, then replace the exit adapters, the full-frame repair, or both with identity. The unchanged trained path and a cached-canvas replay through repair and the output head are checked exactly before scoring. This produces a complete 2×2 intervention with 212 outputs at QP32. The maps contain 403/582/369/411 tiles at depths 6/8/10/12; 45 of the 53 images use at least one early exit. All images, including the eight with only the deepest exit, remain in the analysis.

Removing adapters loses 1.2309 dB RGB PSNR on average, with a paired 95% sequence-bootstrap interval of [0.8765, 1.6582] dB. Removing repair loses 0.00208 dB [0.00060, 0.00363]; removing both loses

1.2538 dB [0.8894, 1.6915] (Fig. S31). These are sensitivities of the trained checkpoint, not comparisons with networks retrained without these components. In particular, the large adapter penalty does not establish that the same quality is unattainable with a different, jointly optimised exit head.

We also partition valid pixels into an 8-pixel-wide band around internal 256-pixel grid boundaries and its complement. Removing repair increases mean within-sequence relative MSE by 0.192% in the band and 0.039% in the interior. Adapter removal changes these by 37.92% and 43.74%, respectively: its effect is not confined to seams. The joint PSNR penalty exceeds the sum of the separate penalties by 0.02085 dB [0.01049, 0.03430] on average; this interaction depends on the dB scale. The small repair benefit motivates an equal-training-budget comparison of repair capacity and end-to-end cost, rather than treating a full-frame repair stage as free.

A uniform-depth context check. On two fixed DIV2K centre768 crops (0801 and 0880), we compare full-frame suffix features with 32-cell cores carrying $k - 4$ feature cells of halo at each depth $k \in \{6, 8, 10, 12\}$. Windows stop at the true image boundary, preserving stock zero padding at every layer. After assembly, both paths use the same adapter/head without repair. At QP32 in CPU FP32, all eight feature and

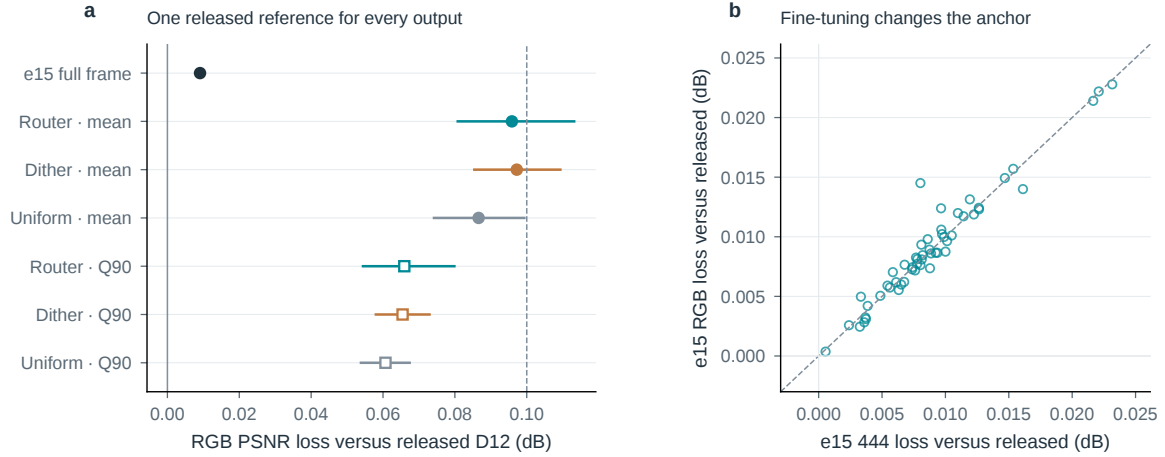


Figure S30. **Measure the anchor before interpreting the loss budget.** The same 53 first frames, QP32, valid-pixel support and CPU FP32 reconstruction. **a**, e15 full-frame and six fixed policies relative to released D12; points and intervals show means and paired 95% sequence-bootstrap intervals. Filled circles use mean control and hollow squares Q90; the dark full-frame point measures anchor drift. The dashed 0.1 dB line is a numerical reference. **b**, Per-sequence e15 drift under equal-channel unclipped 444 and converted/clipped RGB. The diagonal denotes metric agreement. No output or policy is reselected when the reference changes.

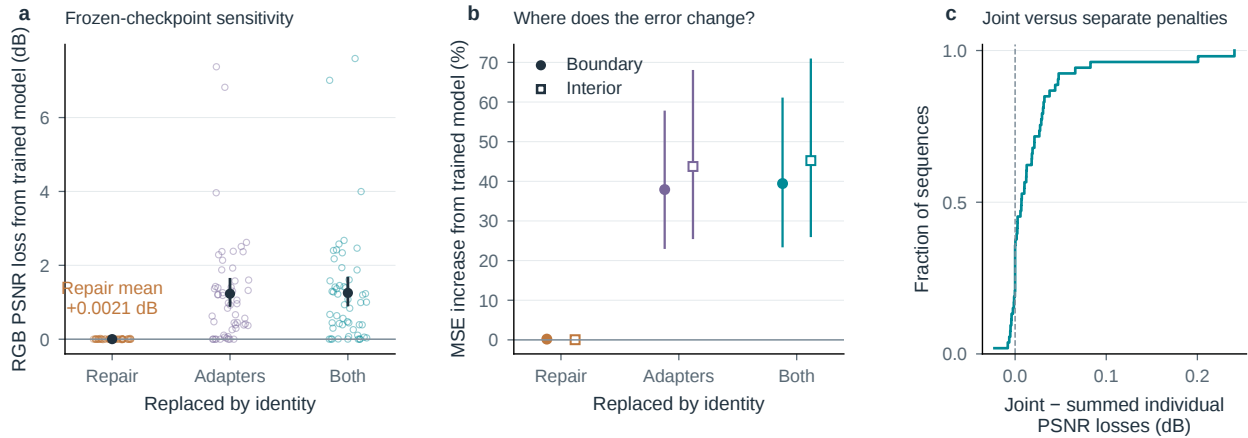


Figure S31. **Separate learned dependence from architectural necessity.** Fixed QP32, checkpoint and Q90 exit maps on all 53 first frames. **a**, RGB PSNR loss after identity substitution: individual sequences, means and paired 95% bootstrap intervals. **b**, Relative MSE change in boundary and interior pixels, averaged within sequence before aggregation. **c**, Joint-minus-summed individual PSNR penalties. This is a frozen-weight sensitivity test; it measures neither retrained alternatives nor runtime.

output comparisons have zero maximum absolute difference. This bounded numerical check supports the context construction; it does not establish mixed-map equivalence, reduced runtime or a quality gain for the deployed replicate-padding and repair path.

S17. The Cost of Independent Spatial Regions

Shared early exit preserves one encoded representation. A bank of complete codecs changes that constraint: independently encoded regions can repeat source context, reset entropy state and introduce additional headers. Before attribut-

ing a gain to expert selection, we measure this cost with the model held fixed.

We predeclare 16 evenly spaced DIV2K centre crops of size 512×512 , five QPs, and three models: D2/D6 at epoch 20 and the released D12 anchor. Each full-frame reconstruction is compared with four independent 256×256 cores, using source-context halos of 0, 32 or 64 pixels. These 240 model-image-QP cases retain every variant. A separate decoder process receives each research bitstream without source pixels; exact reconstruction, latent, quantised-symbol and entropy-index agreement is checked. The CPU FP32

1218 FUFREF1 protocol pads images to multiples of 64. It is not
1219 Microsoft’s native FP16 CUDA wire format.

1220 **Boundary repair can spend more bits than it recovers.**
1221 At 0.2 actual payload bpp, all 16 images are supported by all
1222 four D6 curves. Relative to the same model on the full frame,
1223 independent cores lose 0.0611 dB RGB PSNR; 32- and 64-
1224 pixel halos lose 1.2914 and 1.6666 dB (Fig. S32). These
1225 comparisons interpolate each image in log-rate space on com-
1226 mon measured support, without extrapolation. The respec-
1227 tive 95% paired-image intervals are $[-0.0725, -0.0497]$,
1228 $[-1.4810, -1.1071]$ and $[-1.9113, -1.4178]$ dB. They con-
1229 dition on the small fixed cohort and checkpoint, not training
1230 seeds.

1231 The same-QP diagnostic explains why a visually im-
1232 proved boundary need not improve the rate–distortion com-
1233 parison. At D6/QP32, halo32 changes mean PSNR by only
1234 $+0.0020$ dB, while increasing payload by 39.72% on aver-
1235 age; halo64 changes it by -0.0037 dB and increases payload
1236 by 57.62%. The boundary-specific excess MSE falls towards
1237 zero, but the context is still coded and transmitted (Fig. S33).
1238 This is a joint effect of region support, entropy resets and
1239 redundant context, not an isolated causal estimate for one
1240 component.

1241 For halo32, each 288-square window is padded to 320;
1242 halo64 uses a 320-square window directly. Both therefore
1243 process 1.5625 times the full-frame area, although their ac-
1244 tual source support and coded payload differ. Four FUFREF1
1245 headers occupy 352 bytes versus 88 bytes for the full frame,
1246 an additional 0.00806 bpp at this crop size. At 0.2 complete-
1247 container bpp, the D6 core-only loss is 0.2230 dB on the
1248 15-image common support. The research identity headers
1249 are intentionally explicit and are not a minimum production
1250 signalling format.

1251 The practical implication is to evaluate how regions are
1252 executed alongside which expert is assigned. A shallow
1253 bank cannot recover its expected cost advantage merely by
1254 adding context everywhere. Padding location and coalescing
1255 of adjacent same-depth regions are separate, predeclared
1256 controls; they must preserve the selected pixel depths and
1257 account for all coded bytes.

1258 S17.1. Padding location changes the rate penalty

1259 The first control rounds a 288-square context window to
1260 320 pixels before analysis. We repeat all 240 cases with
1261 a different geometry: keep the image at 288 pixels and
1262 pad its 18-square latent to 20 before hyperanalysis. This
1263 matches the native path’s spatial rounding convention in the
1264 CPU reference implementation. It is not a native CUDA or
1265 FP16 equivalence test. The aligned full-frame, core-only
1266 and halo64 cases remain controls; the full-frame payload
1267 and reconstruction are verified unchanged.

1268 At 0.2 payload bpp, all 16 images support all five variants.

1269 Changing the padding location improves halo32 RGB PSNR
1270 by 0.2791, 0.2652 and 0.2470 dB for D2, D6 and released
1271 D12, respectively (Fig. S35). For D6, the paired 95% interval
1272 is $[0.2354, 0.2951]$ dB. This recovery does not remove the
1273 cost of independently coded context: the corrected halo32
1274 path still trails the same full-frame D6 by 1.0262 dB, with
1275 interval $[0.8614, 1.1915]$. At QP32, the geometry change
1276 reduces payload by 6.03% while changing mean PSNR by
1277 -0.0055 dB. These are distinct comparisons: same-QP byte
1278 reduction and matched-rate quality improvement. Padding
1279 location changes analysis support and hyperprior context
1280 together, so the experiment cannot attribute the effect to
1281 either mechanism alone.

1282 S17.2. Keep the depth map; change how regions are 1283 encoded

1284 We next hold every pixel’s expert assignment fixed while
1285 changing the independent coding regions. The same 16
1286 images and five QPs use D2/D6 at epoch 20 with a 50/50
1287 pixel split. Both complementary phases of vertical halves,
1288 horizontal halves and a checkerboard are retained. Vertical
1289 and horizontal maps each have two implementations: four
1290 independent cores or two coalesced regions. With the two
1291 checkerboard phases, this gives ten profiles per image–QP
1292 pair and 800 checked outputs. The decoder receives only the
1293 fixed-profile research container and resident model weights.

1294 Coalescing removes a coding boundary without reassign-
1295 ing any source pixel. At QP32, mean payload falls by 13.05%
1296 for vertical halves and 12.39% for horizontal halves. Mean
1297 RGB PSNR changes are -0.00045 and -0.00100 dB; their
1298 paired intervals include zero. These small averages do not
1299 establish per-image equivalence. At 0.2 payload bpp, all 16
1300 images are supported, and coalescing improves PSNR by
1301 0.5327 dB $[0.4391, 0.6291]$ for vertical maps and 0.4959 dB
1302 $[0.4121, 0.5748]$ for horizontal maps (Fig. S36). We aver-
1303 age the two phases within each image before bootstrapping
1304 images, rather than counting 32 independent samples.

1305 The complete-container gains at 0.2 bpp are 0.6401 and
1306 0.6084 dB. Our explicit FUFBNK1 framing costs 12 bytes
1307 plus four length bytes per embedded region; each FUFREF2
1308 component also carries an 88-byte header. Thus non-payload
1309 bytes fall from 380 to 196 when four regions become two.
1310 This is a verifiable research format, not a minimal production
1311 format, so its header benefit must remain separate from the
1312 entropy-payload result.

1313 Layer-wise Conv2d tracing gives a 12.44% reduction
1314 in neural-decoder MACs and 11.99% in encoder-plus-
1315 reconstruction MACs for this geometry. These are arithmetic
1316 reductions, not measured latency. Region shape, duplicated
1317 context and entropy resets change together. Connected-
1318 region coding also has prior precedent [1]; the control is
1319 not a standalone novelty claim. Its practical conclusion is
1320 narrower: the same depth decisions can have materially dif-

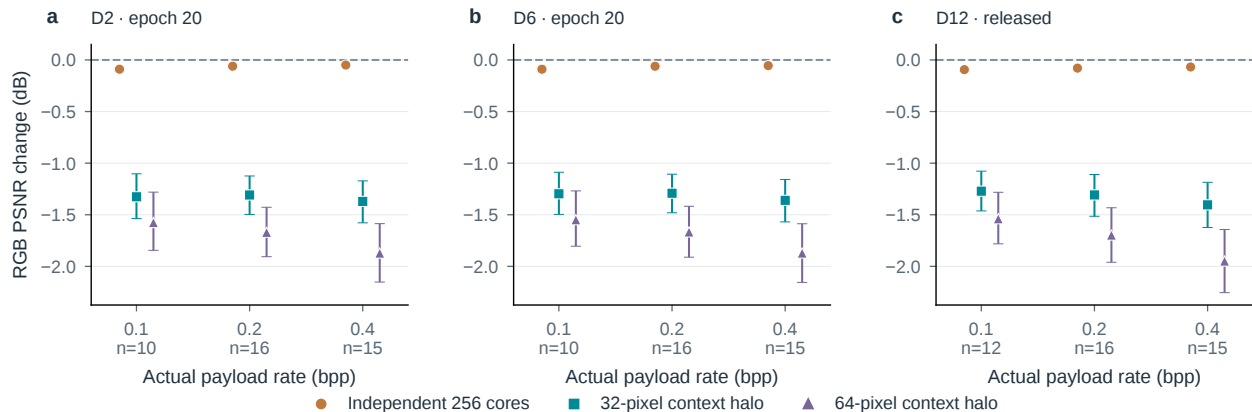


Figure S32. **Charge repeated context to the transmitted rate.** RGB PSNR changes from the same full-frame model at matched actual payload rate. Each depth uses the common supported images across full frame and all three patch variants; counts are shown and the vertical scale is shared. Intervals resample paired images. D2/D6 are interim epoch 20 models; D12 has different released training provenance. The experiment holds depth fixed within every contrast and makes no adaptive-routing or runtime claim.

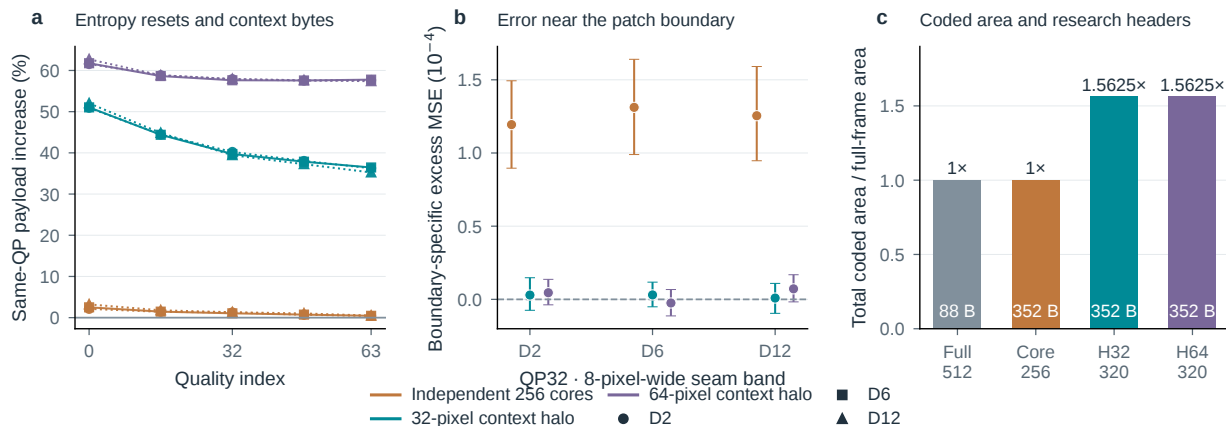


Figure S33. **A better seam and a better codec are different outcomes.** **a**, Same-QP payload increase over the full-frame model. **b**, At QP32, patch-minus-full MSE in the central 8-pixel seam band, minus the corresponding interior excess. **c**, Exact coded area and research-header bytes. Halo32 and halo64 have equal coded area under this padding policy, but different real context. Area ratios are not latency ratios, and same-QP seam diagnostics are not equal-rate quality comparisons.

ferent rate–distortion and compute costs depending on how they are executed.

S17.3. The encoder has a critical path, not a sum of stage costs

The native intra codec also constrains how an adaptive bank should be timed. At the pinned Microsoft revision, analysis and prior processing produce the quantized latent on the main CUDA stream. An event then releases an entropy worker, while synthesis continues on the main stream. The worker gathers symbols, transfers them to the host and performs CPU rANS encoding. The compression call waits for that worker and returns the reconstruction as a GPU tensor; completing the GPU work requires caller-side synchroniza-

tion. Figure S37 records these source-level dependencies. They permit overlap but do not establish how much overlap occurs on a device. Consequently, a sum of isolated stage times is not the encoder’s elapsed time.

For a resident expert bank, the measurement boundary must additionally include source-feature extraction, prediction, grouping, expert dispatch and region assembly. Transfers belong inside that boundary when required by the chosen output contract. Independently trained experts need their own compatible latents; choosing a depth does not make their representations interchangeable. The prepared native correctness harness checks stock versus patched D12 and fresh-process decoding before any GPU latency claim. Its

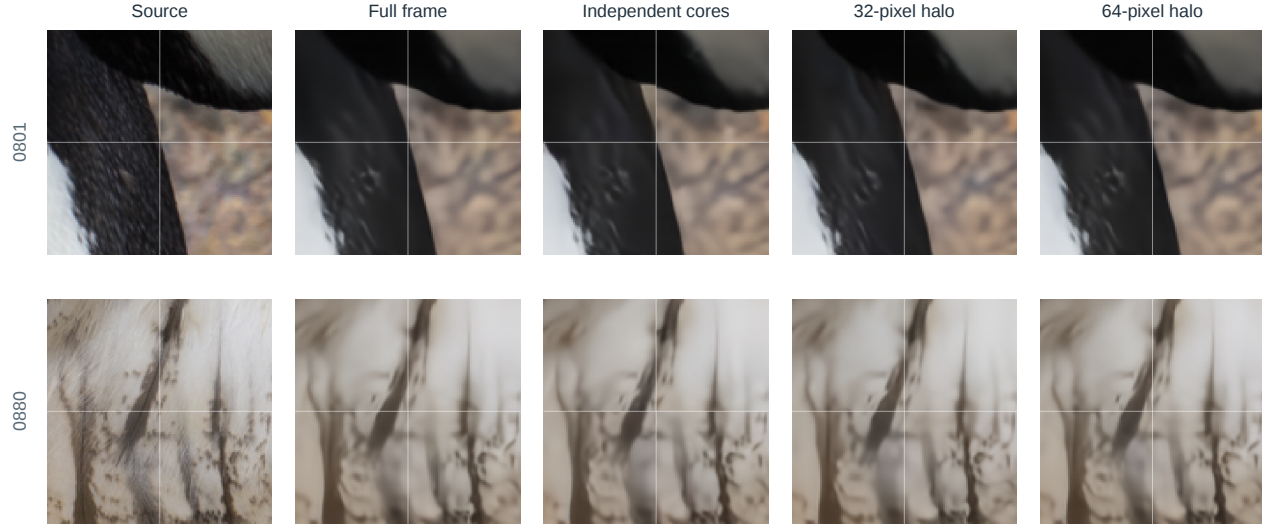


Figure S34. **Predeclared views at the region intersection.** D6 epoch 20, QP32, DIV2K 0801 and 0880. All panels use rows and columns 192:320 of the same 512-square crop; images and window were fixed before outcomes. The common light cross marks the patch boundary. Display images are rounded to 8 bit; numerical metrics use floating-point RGB. These are same-QP views, not equal-bitrate reconstructions.

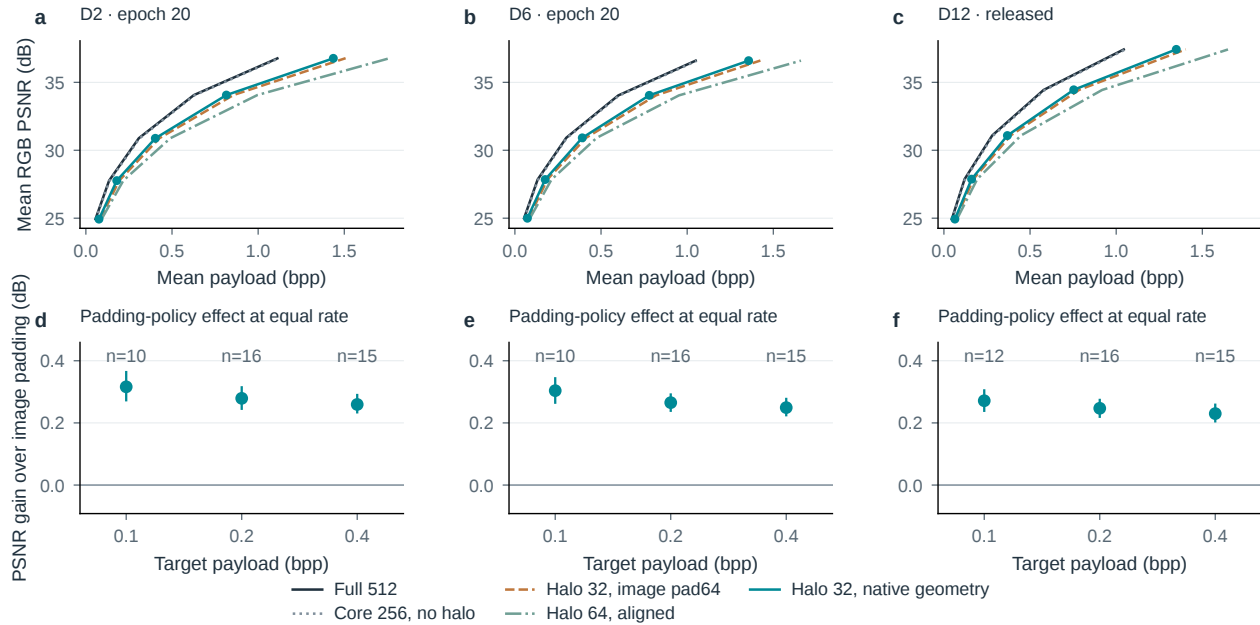


Figure S35. **Keep spatial rounding inside the codec contract.** **a–c,** Actual-payload RD points averaged over the fixed 16-image cohort, with shared axis limits. **d–f,** Per-image matched-rate gain from moving padding to the latent, with paired 95% image-bootstrap intervals. Each rate uses the common support of all five variants within that depth; n denotes supported images out of 16, with no extrapolation. The latent-padding implementation uses CPU FP32 research streams; “native geometry” does not denote native CUDA output or latency.

1347 CPU checkpoint and CDF preflight has passed; its CUDA
1348 execution remains pending.

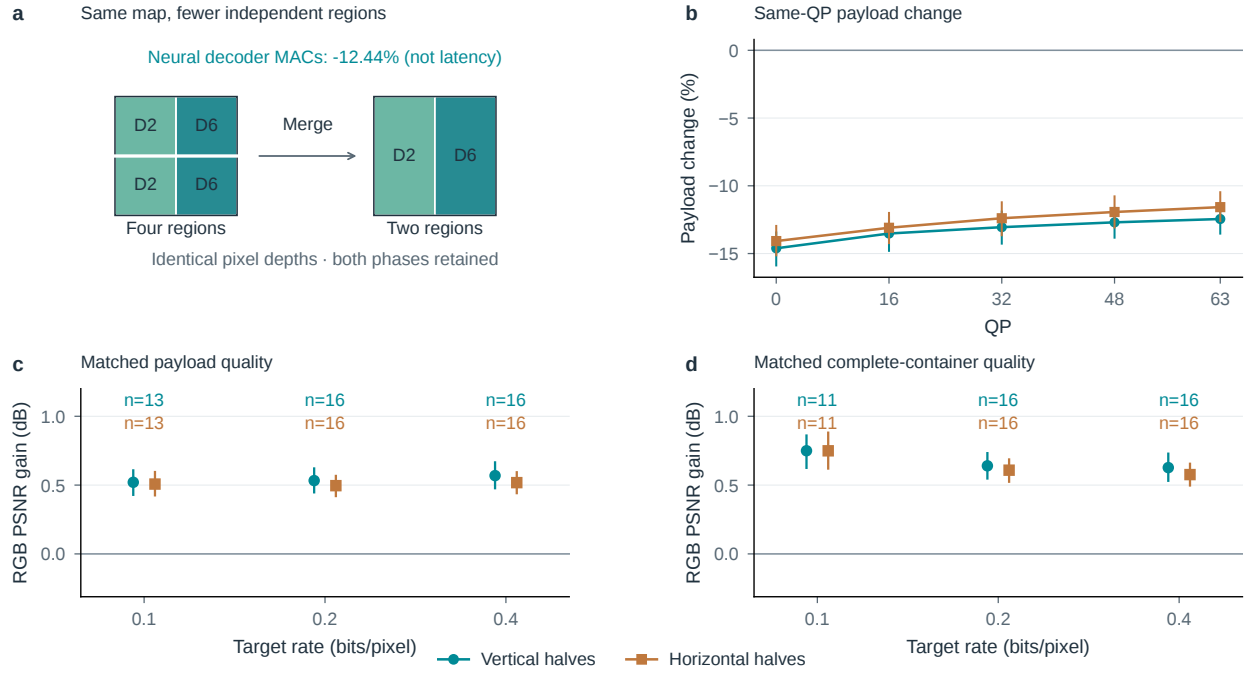


Figure S36. **Better region execution without a better selector.** **a**, One phase of the fixed depth-map control; every pixel keeps its D2 or D6 assignment. **b**, Same-QP relative payload change after coalescing. **c,d**, RGB PSNR gain at matched payload and complete-container rate, on shared vertical scales. Complementary phases are averaged within image before 95% paired-image bootstrap intervals. Counts show the common support of both phases and implementations, without extrapolation. The 800 outputs include unmerged checkerboards; those have no adjacent same-depth regions to combine. CPU FP32 research streams and epoch-20 weights are used throughout; this is neither learned routing nor a runtime benchmark.

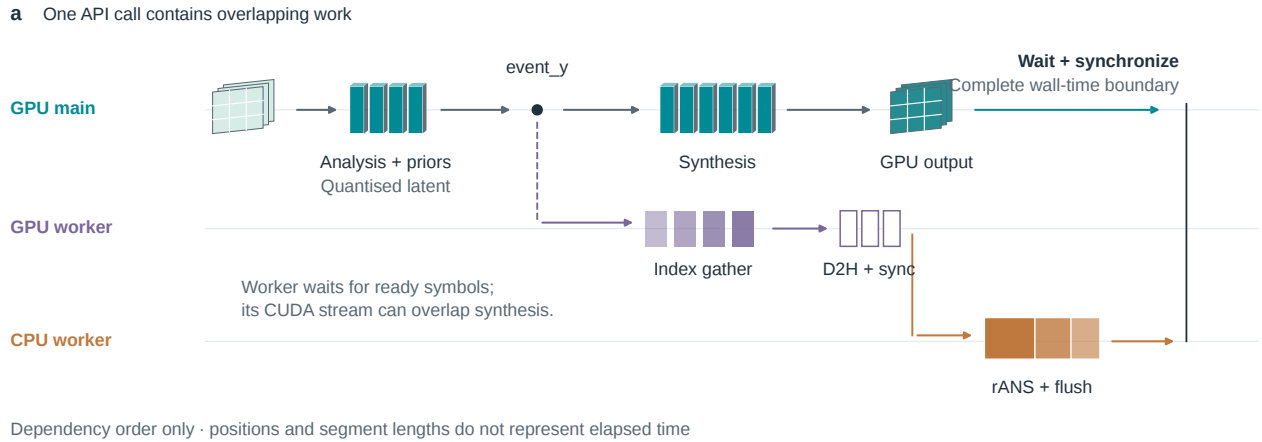


Figure S37. **Time the completed workload across streams.** Source-derived dependencies of the pinned native DCVC-UF intra encoder. The main-stream reconstruction can overlap worker-stream symbol handling and CPU entropy coding after the latent-ready event. The right boundary includes both the worker wait and completion of the GPU reconstruction. Positions, segment lengths and glyph counts are illustrative, not measured durations or layer counts. This diagram establishes a timing contract, not a speedup.

S18. GPU Realization of the Shared Early-Exit Decoder

Question and execution boundary. The arithmetic saving assigned to a spatial exit map becomes useful only if its masked decoder executes efficiently on a GPU. We therefore implemented an opt-in inference path for the existing e15 DCVC-UF shared-exit decoder. It changes neither checkpoint weights nor the recorded exit maps. The six/twelve-block labels below denote exit depths *within that one e15 network*; none of the independently trained D2/D4/D6 codecs is timed or modified here. The measurement boundary starts with the quantized latent and decoder QP already on the GPU and ends when the padded reconstructed tensor is ready. It includes tile selection, shared-prefix execution, adapters, up-sampling, the output block, and grid-seam repair. It excludes analysis-transform encoding, entropy coding/decoding, bit-stream parsing, acquiring the router map, and host-to-device transfer. A synthesis acceleration is thus distinct from a complete decoder or complete codec acceleration.

Operator construction. The dominant DCVC-UF DepthConvBlock contains channel-expanding 1×1 projections followed by weighted SiLU and a four-way channel reduction. The stock PyTorch path materializes the expanded $4C$ -channel tensor and launches separate elementwise and reduction kernels. The Triton path fuses the first projection, activation, and reduction into one kernel that writes only C channels. It also fuses the second projection with its residual addition, and applies analogous pointwise epilogues to the trunk, early-exit adapters, upsample boundary, and RGB head. A dedicated depthwise kernel supports the actual replicate padding used by e15. The full-frame seam module uses a depthwise-activation pass and one fused pointwise-gate-residual pass. The implementation uses $\text{tf}32 \times 3$ matrix multiplication on an RTX A6000 (SM86); inputs and outputs remain FP32. Unsupported layout, dtype, device, or autograd conditions fall back to the stock operators. The wrapper is installed only after checkpoint loading and `eval()`, and converted inference weights are not saved as training checkpoints.

Paired GPU measurements. For videoSRC05 at QP32, we replayed its archived source-calibrated router map, containing 40 tiles and four active depths. Each six-trial block used the same latent, QP, weights, and map, with execution order shuffled and TF32 disabled. CUDA-event medians fell from 282.3 ms for the stock masked path to 254.0 ms after depth-sorted execution, 176.5 ms after activation fusion, 159.3 ms after FFN fusion, 128.2 ms after trunk fusion, and 117.0 ms after adapter fusion. The median of six *paired* stock/fast ratios is $2.40\times$ (individual ratios 2.27–2.46 $\times$). This is a measured synthesis gain at one map and QP, not the

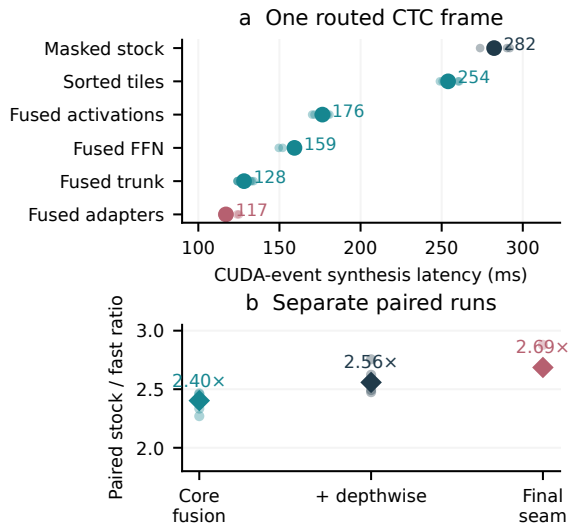


Figure S38. **An implementation gain on one real routed frame.** **a**, Six shuffled, paired CUDA-event trials of consecutive kernel stages. Dots are trials and large marks are medians. **b**, Three separately timed implementations; ratios are paired *within* each run but cannot be compared as an incremental ablation across runs because GPU contention changed. CTC videoSRC05 at QP32, 1920×1080 valid pixels padded to 2048×1280 , with exit-6/8/10/12 tile counts 13/20/4/3. The GPU was shared with training.

average speedup of the CTC corpus.

A subsequent depthwise-enabled version gave 287.2 versus 112.1 ms and a median paired ratio of $2.56\times$ in four further trials on the same frame. The host-wall medians were 289.5 and 114.6 ms, respectively. Adding the seam fusion produced 308.8 versus 113.3 ms in a separate four-trial run. The latter stock measurement drifted upward under shared GPU load; its apparent $2.69\times$ ratio does *not* establish an incremental gain over the preceding version. A direct 16-pair seam-only ablation within the optimized decoder measured 116.8 versus 114.8 ms ($1.035\times$ paired median), a small contribution consistent with the remaining whole-frame work.

Numerical fidelity and memory. The final path was checked against stock masked execution on five real CTC frame/QP/map cases: videoSRC05 at QP 0/32/63, videoSRC01 at QP32, and videoSRC10 at QP32. The largest absolute reconstructed-sample difference was 6.26×10^{-7} ; the largest absolute change in the upstream YUV 6:1:1 PSNR was 2.09×10^{-7} dB. Tile sorting alone was bit-exact; fused dot products are not, owing to FP32 operation order. For the same videoSRC05 map, the incremental active-tensor allocator peak fell from 1,069.5 to 692.1 MB, a 35.3% reduction. This excludes model weights, reserved CUDA memory, and

1423 allocations held by other processes.

1424 **Limits and next benchmark.** These measurements estab-
1425 lish that the early-exit architecture can execute substantially
1426 faster in one GPU synthesis setting, but the A6000 was not
1427 exclusive. Other processes can change clock state, cache
1428 behaviour, and interference between paired blocks. The
1429 53-sequence $\times$ five-QP benchmark is prepared with CUDA-
1430 event and wall-clock timings, checkpoint and source hashes,
1431 and a refusal to run by default when another compute pro-
1432 cess occupies the selected GPU. It has not been executed.
1433 The final publication claim should come from that isolated
1434 benchmark, with sequence-cluster uncertainty and separate
1435 QP slices, followed by an end-to-end bitstream decode that
1436 charges entropy recovery, map signalling, and any online
1437 router cost. The paper bundle contains the six raw GPU
1438 records, a hash manifest, and a CPU audit script reproducing
1439 every number in Fig. S38.

1440 **Why one map is insufficient.** The archived 53-sequence
1441 map inventory already shows a strong QP-dependent work-
1442 load shift: exit 6 receives 79% of selected tiles at QP0 but
1443 37% at QP63; the mean retained trunk depth rises from 6.57
1444 to 8.24 blocks. Different tile occupancies change the number
1445 and shape of pointwise matmuls, gather/scatter traffic, and
1446 shared-prefix reuse. Even at the same pixel count, one dense
1447 depth and four small occupancy groups can have very differ-
1448 ent launch and memory behaviour. The chosen QP32 frame
1449 has 13/20/4/3 tiles at exits 6/8/10/12 and therefore probes
1450 mixed execution, but it is not a representative-weighted CTC
1451 mean by construction. Two small sequences lack archived
1452 maps at QP63; the prepared cohort runner marks their dense
1453 fallback explicitly rather than silently changing the denom-
1454 inator. The paper should report router-map and fallback
1455 groups separately.

1456 **Rejected shortcuts.** Several tempting local optimizations
1457 did not survive the scope check. Pretransposing weights
1458 improved a large common-map timing by only about 0.7%,
1459 with an additional packing cost. A separate pixel-shuffle
1460 kernel saved only 0.024 ms at the $2\times$ stage and had no mea-
1461 surable gain at the $8\times$ output. Smaller pointwise tiles low-
1462 ered register pressure, but did not give stable whole-frame
1463 wins; the chosen FFN and pointwise kernels use approxi-
1464 mately 243 and 251 registers per thread, respectively, without
1465 spills on SM86. Single-pass TF32 shortened isolated 40-tile
1466 pointwise work by about 17–18%, yet increased raw out-
1467 put error to roughly 8×10^{-6} and yielded little gain on the
1468 large common body. We retained $\text{tf32}\times 3$ pending a full
1469 decoder-quality sweep. None of these microbenchmarks was
1470 promoted into the paper’s speedup number.

Deployment accounting. The full codec also performs 1471
entropy recovery and may need to transmit a mode map. An 1472
older QP32 decomposition gave separate medians of about 1473
194.5 ms for entropy and latent preparation and 315.0 ms 1474
for routed synthesis. Applying the one-frame $2.56\times$ syn- 1475
thesis ratio to those older component medians would give 1476
 $(194.5 + 315.0)/(194.5 + 315.0/2.56) = 1.60$ as a hypo- 1477
thetical total decoder factor. Since those components were 1478
recorded at a different time and the kernel ratio comes from 1479
one map, even this Amdahl estimate is not a measured out- 1480
come. Future native timings must hold map acquisition and 1481
bitstream representation fixed, report wall time alongside 1482
CUDA events, and charge the deployment policy for its 1483
actual control overhead. This distinguishes a faster imple- 1484
mentation of shared early exit from a new rate–distortion 1485
operating point. 1486

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